What Do Tabulated Isotherms Certify? Sharp Certain Answers to Mixture-Adsorption Screening Queries

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Abstract*—High-throughput materials discovery for energy-intensive chemical separations queries pure-component adsorption repositories through Ideal Adsorbed Solution Theory (IAST) to screen binary mixture selectivity before synthesis or mixed-gas testing. Existing screening pipelines fit a single parametric isotherm per record or compare a small ensemble of fitted equations; however, because IAST integrates spreading pressure down to zero pressure and evaluates the weaker-adsorbing gas beyond its measured pressure window in 59% of real pairs, curves that agree within measurement tolerance on the observed points diverge sharply at query pressures, capturing only 26% of the certified selectivity range and missing 93.1% of held-out measurements at 5 to $10\times$ extrapolation. We reformulate mixture screening as *certain-answer query processing* over infinite-dimensional nonparametric site-mixture cones, comprising a shape-only concave hinge class $\mathcal{C}_{\mathrm{h}}$ and a completely monotone Langmuir-site Stieltjes class $\mathcal{C}_{\mathrm{L}}$ constrained solely by pointwise measurement tolerance ε and a zero-pressure Henry slope cap $H = \kappa K_{\min}$. We prove an *Envelope Reduction Theorem* that decouples bivariate mixture equilibrium optimization without loss into two univariate spreading-pressure envelopes realized by at most $n+2$ sites per record, and establish two scaling laws: an *Extrapolation-Horizon Law* showing that relative uncertainty diverges at $\rho = \Theta(1/\varepsilon)$ (32,308 evaluations collapse onto a universal master curve with $R^2 = 0.91$), and a *Log-Henry Law* showing that low-pressure slope ambiguity scales as $\Theta(q_1^- \ln H)$. We build IsoCert, a two-phase screening engine that admits records via a misfit-floor linear program and compiles continuum-certified column-generation envelopes offline into compact monotone lookup tables, answering online selectivity-interval queries in $0.15\,\mathrm{ms}$ ($8118\times$ faster than direct LP solving) and threshold queries (`YES`, `NO`, or `UNDETERMINED`) in $25\,\mu\mathrm{s}$ from four table interpolations while computing critical Henry caps $\kappa^\star$ and one-step lookahead measurement advice on demand. Experiments across 13,183 NIST ISODB isotherms and 1,930 real gas pairs demonstrate that IsoCert covers 97.0% of held-out points (≤ 10 bar) and 96.5% at 5 to $10\times$ extrapolation ($1.9\times$ tighter than shape-only envelopes), achieves 100.0% coverage with 0 false decisions across 1,592 certified threshold answers under cap-respecting truth, and resolves 68% of undetermined pairs to target precision in a median of 2 lookahead measurements.

***Index Terms*—Certain answers, incomplete scientific databases, constraint query processing, adsorption isotherm repositories, Ideal Adsorbed Solution Theory, column generation, active data acquisition.**

I. Introduction

Separating chemical mixtures accounts for 10 to 15% of global industrial energy consumption [1], motivating large-scale computational and experimental campaigns to discover porous adsorbents such as metal-organic frameworks (MOFs), covalent organic frameworks (COFs), zeolites, and activated carbons for CO_2 capture, olefin/paraffin splitting, and noble-gas purification [2]–[6]. Because multi-component mixture equilibrium experiments are slow and labor-intensive, community adsorption repositories are overwhelmingly unary: the NIST/ARPA-E Isotherm Database (ISODB) [7] curates thousands of single-component isotherms across 14 benchmark gases, compared to only 272 usable binary mixture series. To bridge this gap, materials discovery pipelines operate as analytical query engines over pure-component tables: given a candidate material's pure-gas records T_A and T_B, a bulk mixture pressure P, and feed composition (y_A, y_B), the engine invokes Ideal Adsorbed Solution Theory (IAST) [8], [9] to evaluate binary selectivity $S(P, y)$ and filter candidates against a target threshold $S(P, y) \geq \tau$.

Standard screening toolkits (`pyIAST` [58], `IAST++` [59], `pyGAPS` [60]) evaluate mixture queries by fitting a single parametric curve to each pure-gas record $T = \{(p_i, q_i)\}_{i=1}^n$, such as Langmuir, dual-site Langmuir (DSL), Sips, or Tóth, or by taking the min-max spread of a small ensemble of fitted families, and then running IAST on the fitted curves as if they were exact over all pressures $p \in [0, \infty)$. Figure 1 exposes why this fit-then-query workflow silently breaks down on real screening queries. In a standard flue-gas query (15:85 CO_2/N_2 at $P = 1$ bar, testing whether $S \geq 50$), CO_2 adsorbs much more strongly than N_2, driving the adsorbed-phase mole fraction of CO_2 to $x \approx 0.982$. By Raoult's law for the adsorbed phase, equality of spreading pressures $\Pi_A(Py_A/x) = \Pi_B(Py_B/(1-x))$ forces the query engine to evaluate the weaker gas N_2 at a hypothetical pure-gas reference pressure $r_B = Py_B/(1-x) \approx 8.3$ bar, which exceeds its highest measured point $P_{\max,B} = 1.01$ bar by more than $8\times$. Across 1,930 real material-condition pairs in ISODB, 59% of queries force the weaker gas above its measured $P_{\max}$. Consequently, finite tables leave two unmeasured functional regimes that directly govern selectivity S. First, above $P_{\max}$, six standard parametric families that agree within 3% inside the measured window diverge by over $2.4\times$ in the unobserved query zone (Fig. 1), causing best-fit point models to miss 93.1% of held-out measurements at 5 to $10\times$ extrapolation. Second, below the lowest measured pressure p_1, spreading pressure $\Pi_q(r) = \int_0^r (q(p)/p)\,dp$ integrates down to zero pressure, so unconstrained behavior on $[0, p_1]$ shifts $\Pi_q(r)$ by an arbitrary additive constant. Across 1,930 real pairs, 6-family ensemble spreads capture a median of only 26% of the certified selectivity width (14.6% of single-model fits fall outside the ensemble), and residual-bootstrap bands remain strictly inside the certified interval on 51% of pairs.

To eliminate uncertified parametric assumptions, we reformulate mixture-adsorption screening as *certain-answer query*

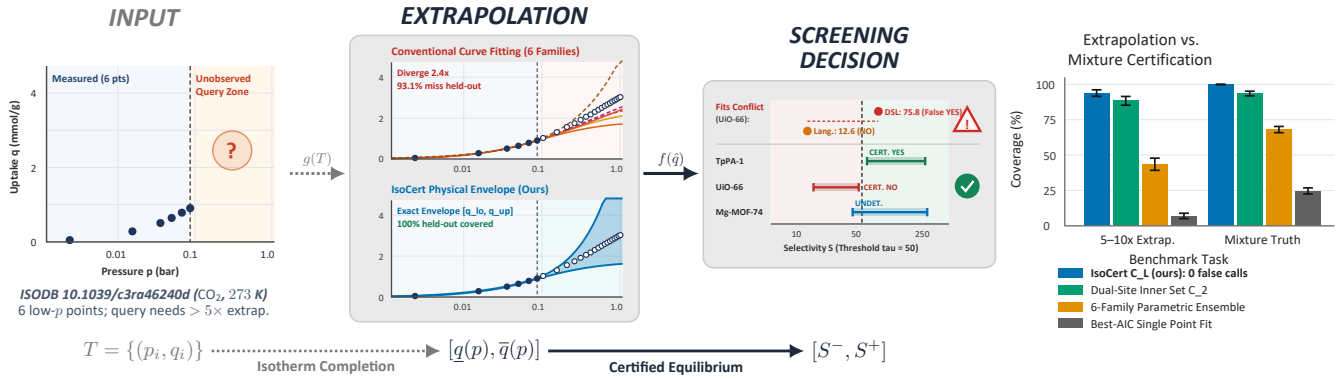


Fig. 1. **Why conventional curve-fitting IAST screening fails and how IsoCert delivers fast, certified certain answers. Left (*Input* → *Extrapolation* → *Screening Decision*):** Given a tabulated pure-gas record $T = \{(p_i, q_i)\}$ (e.g., NIST ISODB record `10.1039/c3ra46240d` with 6 low-pressure points) and a binary mixture query (15:85 CO_2/N_2 at 1 bar, testing threshold $S \geq 50$), IAST equilibrium evaluates the weaker gas more than $5\times$ above its measured window. Six standard parametric families agree closely on the measured points yet fan out by $2.4\times$ in the unobserved query zone, missing 93.1% of held-out measurements and producing contradictory screening calls on `UiO-66` (Dual-Site Langmuir predicts $S = 75.8$, a false `YES`, whereas single-site Langmuir predicts $S = 12.6$, `NO`). In contrast, IsoCert compiles exact nonparametric Stieltjes envelopes $[\underline{q}(p), \overline{q}(p)]$ covering 100% of held-out points on the example curve and decouples online queries into four table lookups (25 μs, $8118\times$ faster than direct LP), separating guaranteed `CERTIFIED YES` (`TpPA-1`: $[56.8, 247.5]$) and `CERTIFIED NO` (`UiO-66`: $[14.7, 46.1]$) answers from `UNDETERMINED` pairs (`Mg-MOF-74`: $[39.5, 260.1]$, resolved via active probing). **Right (*Extrapolation vs. Mixture Certification*):** Across 607 held-out records at 5 to $10\times$ extrapolation and 670 cap-respecting ground-truth mixtures, IsoCert ($\mathcal{C}_{\mathrm{L}}$) achieves 96.5% and 100.0% coverage with **0 false decisions across 1,592 certified threshold answers**, whereas best-AIC point fits drop to 6.9% and 24.6% and 6-family ensembles drop to 58.8% and 58.6%.

processing over incomplete scientific databases [10]–[14]. Rather than guessing a single functional form per table, we model a tabulated record $T = \{(p_i, q_i)\}_{i=1}^{n}$ with pointwise measurement tolerance e_i and a zero-pressure Henry slope cap $q'(0) \leq H = \kappa K_{\min}$ as a finite constraint specification over an infinite-dimensional cone of physically admissible isotherms $\mathcal{C}(T, e, H)$. We study two nested nonparametric functional cones: the *shape-only hinge cone* $\mathcal{C}_{\mathrm{h}}(T, e, H)$ of all nonnegative, nondecreasing, concave isotherms [47], [49] generated by elementary hinge kernels $\phi_b^{\mathrm{h}}(p) = \min(p, 1/b)$, and the *Langmuir-site Stieltjes cone* $\mathcal{C}_{\mathrm{L}}(T, e, H) \subset \mathcal{C}_{\mathrm{h}}(T, e, H)$ of all nonnegative mixtures of elementary Langmuir sites $\phi_b^{\mathrm{L}}(p) = p/(1 + bp)$ over $b \in [0, \infty)$ [31], [34], [37]. Given a pair of records, the certain answer to an interval query is the exact selectivity range $[S^-, S^+]$ across all $(q_A, q_B) \in \mathcal{C}_A \times \mathcal{C}_B$, and the certain answer to a threshold query $S \geq \tau$ is `YES` ($S^- \geq \tau$), `NO` ($S^+ < \tau$), or `UNDETERMINED` ($S^- < \tau \leq S^+$). Because the low-pressure cap $H = \kappa K_{\min}$ cannot be verified from points at $p \geq p_1$ alone, we also elevate κ from a fixed modeling parameter to a queryable output: the *critical Henry cap* $\kappa^\star$, defined as the exact supremum of κ under which a certified `YES` or `NO` decision holds.

Computing exact selectivity bounds over $\mathcal{C}_A \times \mathcal{C}_B$ requires solving a coupled bivariate optimization over two infinite-dimensional functional cones linked through a nonlinear equilibrium fixed point, at sub-millisecond repository query speeds. We resolve this challenge through three theoretical results. First, our **Envelope Reduction Theorem** (Theorem 1) exploits the opposite monotonicity of the Raoult arguments $r_A(x) = Py_A/x$ and $r_B(x) = Py_B/(1-x)$ to prove that bivariate optimization over $\mathcal{C}_A \times \mathcal{C}_B$ decouples *without relaxation loss* into two univariate spreading-pressure envelopes per record, $U_g(r) = \sup_{q \in \mathcal{C}_g} \Pi_q(r)$ and $L_g(r) = \inf_{q \in \mathcal{C}_g} \Pi_q(r)$, each attained by a discrete mixture of at most $n+2$ sites (Theorem 2). This decoupling reduces any threshold query $S \geq \tau$ to four univariate envelope evaluations at x_τ (Corollary 1). Second, we establish **Law 1 (Extrapolation-Horizon Law, Theorem 3)**, proving via an explicit two-site witness that relative uptake and spreading-pressure uncertainty remain $O(\varepsilon)$ inside the measured window but diverge at the dimensionless horizon $\rho = r/P_{\max} \sim (1+1/c)/\varepsilon$, which collapses 32,308 evaluations across 1,542 records onto a single master curve ($R^2 = 0.91$). Third, we establish **Law 2 (Log-Henry Law, Theorem 4)**, proving that low-pressure spreading-pressure width on $[0, p_1]$ scales logarithmically with the Henry cap as $\Theta(q_1^- \ln H)$, explaining why an explicit slope cap is both necessary and sufficient for finite certification.

Guided by these theoretical results, we build **IsoCert** (Section V, Fig. 2), a two-phase compilation, query, and active acquisition engine for isotherm repositories. Offline, IsoCert solves a misfit-floor linear program (LP) to admit records ($\varepsilon^\star \leq 3\%$) and compiles continuum-certified spreading-pressure envelopes $(U_g, L_g, q_{\mathrm{lo},g})$ on a 96-node geometric pressure grid using column generation over site affinities $b \in [0, \infty)$, combining interval bisection for interior dual violations with an analytic high-b tail repair (Lemma 4) and monotone outer grid interpolation (Proposition 1). Online, IsoCert answers threshold queries in 25 μs from four table interpolations and interval queries in 0.15 ms via bracketed scalar root-finding ($8118\times$ faster than direct LP evaluation at 1235 ms), pairs its outer certificate with a parametric dual-site inner solver ($\mathcal{C}_2$) realizing 96% of the certified width, and guides one-step lookahead experimental acquisition when a query is `UNDETERMINED`. Our contributions are summarized as follows:

- **Certain-Answer Semantics and Exact Envelope Decoupling (Sections III and IV-A):** We formulate mixture screening over incomplete pure-gas tables as certain-answer query processing over nonparametric hinge ($\mathcal{C}_{\mathrm{h}}$) and Langmuir-site ($\mathcal{C}_{\mathrm{L}}$) cones, and prove that bivariate selectivity optimization decouples without loss into univariate $(n+2)$-site spreading-pressure envelopes (Theorems 1

and 2, Corollary 1).

- **Fundamental Certification Laws (Sections IV-B and IV-C):** We establish the *Extrapolation-Horizon Law* ($\rho = \Theta(1/\varepsilon)$, Theorem 3) and the *Log-Henry Law* ($\Delta\Pi(p_1) = \Theta(q_1^- \ln H)$, Theorem 4), characterizing the exact regimes where high-pressure tail divergence and low-pressure slope ambiguity govern query precision.
- **Two-Phase Compilation, Query, and Active Acquisition Engine (Section V):** We design IsoCert, integrating continuum-certified column generation (Lemma 4), monotone outer grid compilation (Proposition 1), $25\,\mu$s threshold filtering, critical Henry cap bisection, dual-site inner bracketing, and one-step lookahead experimental acquisition (Algorithm 1).
- **Repository-Scale Evaluation on NIST ISODB (Section VI):** Across 13,183 isotherms and 1,930 real gas pairs, IsoCert admits 68.5% of records (Table I), covers 97.0% of held-out measurements overall and 96.5% at 5 to $10\times$ extrapolation while cutting shape-only width by $1.9\times$ (Table II), achieves 100.0% ground-truth coverage with **0 false decisions across 1,592 certified threshold queries** (Table III), certifies 66% `YES` and 50% `NO` calls on real pairs at a $2\times$ margin (Table IV), and resolves 68% of undetermined pairs in a median of 2 lookahead measurements (Fig. 5).

II. Related Work

Answering queries over incomplete records by intersecting query results across all completions (*possible worlds*) consistent with integrity constraints has been a foundational theme in database theory and systems since Imieliński and Lipski [10], Abiteboul *et al.* [11], and Libkin [12], [13], [16], [17]. Constraint databases [14], [15], [18] generalize finite relational tuples to semi-algebraic or linear constraint regions, whereas uncertain and probabilistic database systems such as Trio [22], MayBMS [23], [24], MystiQ [19], [20], and range or package query engines [21], [25], [26] propagate tuple-level or attribute-level uncertainty through relational operators, and active query systems [27]–[30] acquire missing values to resolve ambiguous answers. Our work extends certain-answer query processing to *continuous scientific functional tables*, where each record is a finite pressure-uptake sample constraining an infinite-dimensional cone of physical functions and the query operator is a nonlinear thermodynamic equilibrium fixed point. Whereas general bivariate functional optimization over $\mathcal{C}_A \times \mathcal{C}_B$ would require joint optimization at query time, our Envelope Reduction Theorem (Theorem 1) acts as an *exact functional projection* that compiles each unary record independently into two univariate envelope tables offline, enabling $25\,\mu$s online threshold certification and active lookahead acquisition.

Our Langmuir-site cone $\mathcal{C}_{\mathrm{L}}$ builds on the classical representation of completely monotone and Bernstein functions as Stieltjes transforms of nonnegative measures [31]–[34], which unifies heterogeneous adsorption models (Langmuir [56], dual-site Langmuir [57], Tóth [36], and UNILAN [37]–[39]) while excluding models with unphysical infinite zero-pressure slope such as Sips ($\nu < 1$) [35]. Bounding linear functionals such as uptake $q(r)$ or spreading pressure $\Pi_q(r)$ over a measure cone subject to finitely many interval constraints is a Chebyshev moment and linear semi-infinite programming (LSIP) problem [40]–[46], serving as the optimal-recovery [51]–[53] and interval-analysis [54], [55] counterpart to nonparametric shape-constrained regression [47]–[50]. Unlike standard LSIP discretizations that solve on a fixed grid without guaranteeing validity between grid nodes, IsoCert combines interval bisection with an analytic high-affinity tail repair (Lemma 4) and outer monotone grid envelopes (Proposition 1) to guarantee continuum soundness across all $b \in [0, \infty)$ and $r > 0$, while establishing closed-form scaling laws for the extrapolation horizon (Theorem 3) and low-pressure Henry cap (Theorem 4).

In chemical engineering and materials informatics, the NIST/ARPA-E ISODB repository [7] and porous-material structural databases [4]–[6] have enabled repository-scale screening via IAST packages such as `pyIAST` [58], `IAST++` [59], and `pyGAPS` [60], alongside interlaboratory reproducibility benchmarks [64]–[72]. Recent studies propagate isotherm uncertainty through IAST using Bayesian parameter posteriors [61], local parametric sensitivity [62], [63], residual bootstrap resampling, or small multi-equation ensembles. Because these methods restrict the completion set to one or a few low-dimensional parametric equations and rely on implicit tail extrapolations above $P_{\max}$, they miss 56.5% of held-out measurements at 5 to $10\times$ extrapolation and capture only 26% of the certified selectivity width across 1,930 real pairs. IsoCert replaces uncertified parametric spreads with deterministic certain answers over nonparametric physical cones.

III. Problem Formulation

As illustrated in Fig. 2, IsoCert transforms incomplete pure-component isotherm tables into certified mixture-screening answers by combining physical site-mixture completion cones (Section III-C) with univariate envelope decoupling (Section IV) and two-phase compilation and active acquisition (Section V). We begin by formalizing pure-component isotherm records, the IAST equilibrium operator, our nested completion cones, and three-valued certain-answer semantics.

A. Tabulated Isotherm Records and Screening Queries

Definition 1 (Tabulated Isotherm Record)**.** *A pure-component isotherm record for gas g on a porous material at fixed temperature is an ordered table of n pressure and uptake pairs*

$$T = \{(p_i, q_i)\}_{i=1}^{n}, \quad 0 < p_1 < p_2 < \cdots < p_n = P_{\max}, \quad (1)$$

where p_i denotes bulk gas pressure (bar) and $q_i \geq 0$ denotes gravimetric or volumetric equilibrium uptake ($\mathrm{mmol/g}$), accompanied by a pointwise measurement tolerance vector $e = (e_1, \ldots, e_n) \in \mathbb{R}^n_{>0}$. Following interlaboratory repeatability studies on NIST ISODB [66]–[68], our default relative tolerance is $e_i = \varepsilon \max_{1 \leq j \leq n} q_j$ with $\varepsilon = 0.03$ (3% of maximum uptake).

We write $q_i^- = \max(0, q_i - e_i)$ and $q_i^+ = q_i + e_i$ for the admissible uptake interval at each measured pressure p_i. Every concave isotherm passing through these data boxes satisfies $q(0) = 0$ and must have a zero-pressure Henry slope $q'(0) = \lim_{p \to 0^+} q(p)/p$ at least as large as the steepest lower secant from the origin:

$$K_{\min}(T, e) = \max_{1 \leq i \leq n} \frac{q_i^-}{p_i}. \quad (2)$$

Here $K_{\min}(T,e)$ represents the minimum zero-pressure affinity forced by the observed uptake boxes. Because no measurements exist on the open interval $(0,p_1)$, an unconstrained curve could rise arbitrarily steeply near $p=0$ and flatten before p_1, making the spreading-pressure integral unbounded. Each record completion is therefore parameterized by an explicit *Henry slope cap* $q'(0) \le H = \kappa K_{\min}(T,e)$ with relative multiplier $\kappa \ge 1$ (default $\kappa = 10$).

Given two records T_A and T_B measured on the same material at the same temperature for gases A and B, a mixture screening query specifies bulk pressure $P > 0$ and feed mole fractions $y = (y_A, y_B)$ with $y_A + y_B = 1$ ($y_A, y_B \in (0,1)$). We support two fundamental query types: a *Selectivity Interval Query* $\mathrm{Q}_{\mathrm{int}}(P,y)$, which returns the tightest certified interval $[S^-, S^+]$ guaranteed to contain the true binary mixture selectivity $S(P,y)$, and a *Selectivity Threshold Query* $\mathrm{Q}_{\mathrm{thr}}(P,y,\tau)$, which decides for a target threshold $\tau > 0$ whether $S(P,y) \ge \tau$ holds across all admissible completions (`YES`), fails across all completions (`NO`), or cannot be decided from the existing tables (`UNDETERMINED`).

B. IAST Equilibrium as a Functional Database Operator

Ideal Adsorbed Solution Theory (IAST) [8], [9] predicts binary mixture equilibrium from two continuous pure-component isotherms $q_A, q_B : [0,\infty) \to [0,\infty)$ satisfying $q_g(0) = 0$ and finite Henry slope $q_g'(0) < \infty$. To couple the two pure isotherms thermodynamically, IAST evaluates the reduced *spreading pressure* at hypothetical pure-gas reference pressure $r \ge 0$ via the Stieltjes-type integral from zero pressure:

$$\Pi_q(r) = \int_0^r \frac{q(p)}{p}\, dp. \tag{3}$$

Because the integrand $q(p)/p$ weights low pressures inversely by p, behavior near $p = 0$ directly shifts $\Pi_q(r)$ at all higher pressures. At bulk pressure P and gas-phase composition (y_A, y_B), let $x \in (0,1)$ denote the adsorbed-phase mole fraction of component A, leaving $1-x$ for component B. By Raoult's law for an ideal adsorbed phase, the hypothetical pure-gas reference pressures of the two species are

$$r_A(x) = \frac{P y_A}{x}, \qquad r_B(x) = \frac{P y_B}{1-x}. \tag{4}$$

Notice that as component A dominates the adsorbed phase ($x \to 1$), $r_A(x)$ approaches the partial pressure $P y_A$ while the weaker component's reference pressure $r_B(x)$ grows as $P y_B/(1-x) \gg P$. Thermodynamic equilibrium requires equality of spreading pressures between the two components:

$$\Pi_{q_A}\left(\frac{P y_A}{x}\right) = \Pi_{q_B}\left(\frac{P y_B}{1-x}\right). \tag{5}$$

When q_A and q_B are not identically zero, $\Pi_{q_A}(r_A(x))$ is continuous and strictly decreasing in $x \in (0,1)$ (from ∞ as $x \to 0^+$ to $\Pi_{q_A}(P y_A) > 0$ as $x \to 1^-$), whereas $\Pi_{q_B}(r_B(x))$ is continuous and strictly increasing in $x \in (0,1)$ (from $\Pi_{q_B}(P y_B) > 0$ at $x \to 0^+$ to ∞ at $x \to 1^-$). Consequently, Eq. (5) admits a unique root $x^\star = X(\Pi_{q_A}, \Pi_{q_B}) \in (0,1)$, which determines the binary adsorption selectivity of gas A over gas B:

$$S(q_A, q_B; P, y) = \frac{x^\star / y_A}{(1-x^\star)/y_B} = \frac{y_B}{y_A} \cdot \frac{x^\star}{1-x^\star}. \tag{6}$$

Crucially, S is a strictly increasing bijection of $x^\star \in (0,1)$ and depends on q_A and q_B *solely through their spreading-pressure functions* Π_{q_A} and Π_{q_B}. Total mixture uptake q_{tot} then follows from Amagat's law, $q_{\mathrm{tot}}^{-1} = x^\star/q_A(r_A(x^\star)) + (1-x^\star)/q_B(r_B(x^\star))$.

C. Nonparametric Physical Completion Cones

Rather than guessing a single parametric equation per record, we model the set of possible worlds for (T,e,H) as a convex cone of functions generated by physically meaningful site kernels.

1) Shape-Only Concave Hinge Cone ($\mathcal{C}_{\mathrm{h}}$): The weakest thermodynamic shape restriction for Type-I single-layer or micropore adsorption [69], [70] requires $q : [0,\infty) \to [0,\infty)$ to satisfy $q(0) = 0$ and be nondecreasing and concave. Every such function is the uniform limit of nonnegative combinations of a linear Henry term $c_0 p$ and elementary *hinge sites* $\phi_b^{\mathrm{h}}(p) = \min(p, 1/b)$ indexed by saturation reciprocal $b > 0$:

$$q(p) = c_0 p + \int_{(0,\infty)} \min\left(p, \frac{1}{b}\right) d\mu(b), \quad c_0 \ge 0,\ \mu \ge 0. \tag{7}$$

Here μ is a nonnegative Borel measure over site affinities $b \in (0,\infty)$, the zero-pressure slope is $q'(0) = c_0 + \int_{(0,\infty)} d\mu(b)$, and each hinge site integrates via Eq. (3) to the closed-form spreading-pressure kernel

$$\pi_b^{\mathrm{h}}(r) = \int_0^r \frac{\min(p, 1/b)}{p}\, dp = \begin{cases} r, & r \le 1/b, \\ \frac{1}{b}\big(1 + \ln(br)\big), & r > 1/b. \end{cases} \tag{8}$$

2) Langmuir-Site Stieltjes Cone ($\mathcal{C}_{\mathrm{L}}$): Statistical mechanics of independent binding sites and heterogeneous patchwise surfaces [37], [38], [56] represents an isotherm as a nonnegative mixture of elementary *Langmuir sites* $\phi_b^{\mathrm{L}}(p) = p/(1+bp)$ weighted by a nonnegative affinity distribution μ plus an unsaturating linear Henry component $c_0 p$:

$$q(p) = c_0 p + \int_{(0,\infty)} \frac{p}{1+bp}\, d\mu(b), \quad c_0 \ge 0,\ \mu \ge 0. \tag{9}$$

Functions of the form (9) are precisely the Bernstein functions $p f(p)$ where f is a Stieltjes transform [31], [34]; they satisfy $(-1)^{k-1} q^{(k)}(p) \ge 0$ for all $k \ge 1$, meaning every smooth derivative alternates in sign. Substituting $\phi_b^{\mathrm{L}}(p)$ into Eq. (3) yields the exact Langmuir spreading-pressure kernel

$$\pi_b^{\mathrm{L}}(r) = \int_0^r \frac{dp}{1+bp} = \frac{\ln(1+br)}{b}, \qquad \pi_0^{\mathrm{L}}(r) = r. \tag{10}$$

Because every Langmuir site is concave, $\mathcal{C}_{\mathrm{L}}(T,e,H) \subset \mathcal{C}_{\mathrm{h}}(T,e,H)$ strictly. Moreover, classical single-site Langmuir ($\mu = q_{\mathrm{sat}} b_0 \delta_{b_0}$), m-site Langmuir ($\mu = \sum_{k=1}^m q_k b_k \delta_{b_k}$), Tóth [36], and UNILAN [37] isotherms with finite Henry constants all belong to $\mathcal{C}_{\mathrm{L}}$.

Definition 2 (Admissible Record Completions)**.** *For class $\mathcal{C} \in \{\mathcal{C}_{\mathrm{h}}, \mathcal{C}_{\mathrm{L}}\}$, the completion set $\mathcal{C}(T,e,H)$ consists of all functions q of the form* (7) *or* (9) *satisfying the n data-box constraints and the zero-pressure Henry cap:*

$$q_i^- \le q(p_i) \le q_i^+, \qquad \forall i \in \{1,\dots,n\}, \tag{11}$$

$$q'(0) = c_0 + \int_{(0,\infty)} d\mu(b) \le H. \tag{12}$$

3) Parametric Dual-Site Inner Subset ($\mathcal{C}_2$): To bracket the infinite-dimensional cone $\mathcal{C}_{\mathrm{L}}(T,e,H)$ from the inside, we also define the 4-parameter dual-site Langmuir subset $\mathcal{C}_2(T,e,H) = \{q(p) = \sum_{k=1}^{2} \frac{a_k p}{1+b_k p} : a_k \geq 0, b_k \geq 0, \text{Eqs. (11) and (12) hold}\} \subset \mathcal{C}_{\mathrm{L}}(T,e,H)$. Any extremum over $\mathcal{C}_2$ is a valid *inner bound* realized by a smooth two-site isotherm, whereas any extremum over $\mathcal{C}_{\mathrm{L}}$ or $\mathcal{C}_{\mathrm{h}}$ is a valid *outer certificate*, forming the nested hierarchy

$$\mathcal{C}_2(T,e,H) \subseteq \mathcal{C}_{\mathrm{L}}(T,e,H) \subset \mathcal{C}_{\mathrm{h}}(T,e,H). \tag{13}$$

D. Certain-Answer Semantics and the Critical Henry Cap

Given completion cones $\mathcal{C}_A = \mathcal{C}(T_A,e_A,H_A)$ and $\mathcal{C}_B = \mathcal{C}(T_B,e_B,H_B)$, the exact certain answer to $\mathrm{Q}_{\mathrm{int}}(P,y)$ is the selectivity interval $[S^-,S^+]$, where the endpoints are defined by optimizing Eq. (6) over the product of completion cones:

$$S^- = \inf_{\substack{q_A\in\mathcal{C}_A\\ q_B\in\mathcal{C}_B}} S(q_A,q_B;P,y), \quad S^+ = \sup_{\substack{q_A\in\mathcal{C}_A\\ q_B\in\mathcal{C}_B}} S(q_A,q_B;P,y). \tag{14}$$

Accordingly, the three-valued certain answer to the threshold query $\mathrm{Q}_{\mathrm{thr}}(P,y,\tau)$ is

$$\mathrm{Ans}(P,y,\tau) = \begin{cases} \texttt{YES}, & \text{if } S^- \geq \tau, \\ \texttt{NO}, & \text{if } S^+ < \tau, \\ \texttt{UNDETERMINED}, & \text{if } S^- < \tau \leq S^+. \end{cases} \tag{15}$$

Finally, viewing $S^-(\kappa)$ and $S^+(\kappa)$ as functions of the common relative Henry multiplier $\kappa = H_g/K_{\min}(T_g,e_g) \geq 1$, the *critical Henry cap* $\kappa^\star(\tau)$ is the supremum of $\kappa \geq 1$ for which $\mathrm{Ans}(P,y,\tau) \in \{\texttt{YES},\texttt{NO}\}$, quantifying the exact low-pressure regularity margin of a certified screening decision.

IV. Envelope Reduction and Certification Laws

Solving Eq. (14) directly appears to require joint optimization over pairs of infinite-dimensional measures (μ_A,μ_B) coupled through the implicit equilibrium root of Eq. (5). In this section, we first prove that the bivariate equilibrium problem decouples *without loss* into univariate spreading-pressure envelopes supported on at most $n+2$ sites per record (Section IV-A). We then establish two fundamental laws that govern the width of those envelopes: the *Extrapolation-Horizon Law* above $P_{\max}$ (Section IV-B) and the *Log-Henry Law* below p_1 (Section IV-C).

A. Exact Bivariate-to-Univariate Envelope Reduction

To decouple the two gases in Eq. (14), we define for each record completion set $\mathcal{C}_g$ ($g \in \{A,B\}$) and reference pressure $r \geq 0$ its *upper and lower spreading-pressure envelopes*:

$$U_g(r) = \sup_{q\in\mathcal{C}_g} \Pi_q(r), \qquad L_g(r) = \inf_{q\in\mathcal{C}_g} \Pi_q(r), \tag{16}$$

together with the lower uptake envelope $q_{\mathrm{lo},g}(r) = \inf_{q\in\mathcal{C}_g} q(r)$. Our decoupling rests on three structural lemmas establishing regularity, monotone root shift, and compact attainment.

Lemma 1 (Well-Posed Envelopes)**.** *Suppose $\mathcal{C}(T,e,H)$ is nonempty and $q_i^- > 0$ for at least one $i \in \{1,\dots,n\}$. Then for every $r > 0$, $0 < L(r) \leq U(r) \leq Hr < \infty$, and both $U(r)$ and $L(r)$ are continuous and strictly increasing on $(0,\infty)$ with $\lim_{r\to 0^+} U(r) = \lim_{r\to 0^+} L(r) = 0$ and $\lim_{r\to\infty} L(r) = \infty$.*

Proof sketch: Upper boundedness $U(r) \leq Hr$ follows from $q(p) \leq Hp$. Because some $q_i^- > 0$, concavity forces $q(p) \geq q_i^- \min(p/p_i, 1)$ for all $q \in \mathcal{C}$, so $L(r) \geq q_i^- \int_0^r \min(1/p_i, 1/p)\,dp > 0$, which diverges as $\Omega(\ln r)$ as $r \to \infty$. Strict monotonicity on (r_1,r_2) follows from $q(p) \geq q_{\mathrm{lo}}(r_1) > 0$ on $[r_1,r_2]$, and continuity follows from the uniform Lipschitz bound $|U(r_2) - U(r_1)| \leq H|r_2 - r_1|$.

Lemma 2 (Monotone Equilibrium Root Shift)**.** *Let $\Pi_A, \widetilde{\Pi}_A, \Pi_B, \widetilde{\Pi}_B : (0,\infty) \to (0,\infty)$ be continuous, strictly increasing functions that vanish at 0 and diverge at ∞. If $\Pi_A(r) \leq \widetilde{\Pi}_A(r)$ and $\Pi_B(r) \geq \widetilde{\Pi}_B(r)$ for all $r > 0$, then their IAST equilibrium roots satisfy $X(\Pi_A,\Pi_B) \leq X(\widetilde{\Pi}_A,\widetilde{\Pi}_B)$.*

Proof sketch: Define $F(x) = \Pi_A(Py_A/x) - \Pi_B(Py_B/(1-x))$ and $\widetilde{F}(x) = \widetilde{\Pi}_A(Py_A/x) - \widetilde{\Pi}_B(Py_B/(1-x))$. Both F and $\widetilde{F}$ are continuous and strictly decreasing on $(0,1)$, and pointwise dominance gives $F(x) \leq \widetilde{F}(x)$ for all $x \in (0,1)$. At $x^\star = X(\Pi_A,\Pi_B)$, $\widetilde{F}(x^\star) \geq F(x^\star) = 0$, so the unique zero of $\widetilde{F}$ lies in $[x^\star,1)$.

Lemma 3 (Convexity and Compact Attainment)**.** *For $\mathcal{C} \in \{\mathcal{C}_{\mathrm{h}}(T,e,H), \mathcal{C}_{\mathrm{L}}(T,e,H)\}$, the set $\mathcal{C}$ is convex and weak-$\star$ compact, and the map $q \mapsto \Pi_q(r)$ is linear and continuous for every $r > 0$. Consequently, the supremum $U(r)$ and infimum $L(r)$ in Eq. (16) are attained by isotherms $q_r^{\mathrm{up}}, q_r^{\mathrm{lo}} \in \mathcal{C}$.*

Combining Lemmas 1 to 3 yields our first main theorem: bivariate optimization over $\mathcal{C}_A \times \mathcal{C}_B$ reduces *without relaxation loss* to two univariate root solves on the precomputed envelopes (U_A,L_A) and (U_B,L_B).

Theorem 1 (Envelope Reduction Theorem)**.** *Let $\mathcal{C}_A$ and $\mathcal{C}_B$ be nonempty completion sets satisfying Lemma 1. For any bulk pressure $P > 0$ and composition $y = (y_A,y_B)$, let $x^- = X(L_A,U_B)$ and $x^+ = X(U_A,L_B)$ be the unique roots in $(0,1)$ of the univariate envelope equations*

$$\begin{aligned} L_A(Py_A/x^-) &= U_B\big(Py_B/(1-x^-)\big), \\ U_A(Py_A/x^+) &= L_B\big(Py_B/(1-x^+)\big). \end{aligned} \tag{17}$$

Then the exact selectivity bounds in Eq. (14) are attained and given by

$$S^- = \frac{y_B}{y_A}\frac{x^-}{1-x^-}, \qquad S^+ = \frac{y_B}{y_A}\frac{x^+}{1-x^+}. \tag{18}$$

Proof: For every pair $(q_A,q_B) \in \mathcal{C}_A \times \mathcal{C}_B$, definition (16) implies $L_A(r) \leq \Pi_{q_A}(r) \leq U_A(r)$ and $L_B(r) \leq \Pi_{q_B}(r) \leq U_B(r)$ for all $r > 0$. Applying Lemma 2 gives $x^- = X(L_A,U_B) \leq X(\Pi_{q_A},\Pi_{q_B}) \leq X(U_A,L_B) = x^+$, proving that Eq. (18) is a valid outer bracket. Conversely, at the specific numbers $r_A^+ = Py_A/x^+$ and $r_B^+ = Py_B/(1-x^+)$, Lemma 3 guarantees single isotherms $q_A^\star \in \mathcal{C}_A$ and $q_B^\star \in \mathcal{C}_B$ achieving $\Pi_{q_A^\star}(r_A^+) = U_A(r_A^+)$ and $\Pi_{q_B^\star}(r_B^+) = L_B(r_B^+)$. For this valid pair $(q_A^\star,q_B^\star) \in \mathcal{C}_A \times \mathcal{C}_B$, Eq. (5) holds at $x = x^+$, so $X(\Pi_{q_A^\star},\Pi_{q_B^\star}) = x^+$ and $S(q_A^\star,q_B^\star;P,y) = S^+$. An identical construction at (r_A^-,r_B^-) attains S^-. No single isotherm q_A needs to equal $U_A(r)$ at all pressures simultaneously: the equilibrium root x^+ evaluates the envelope at only one pressure $r_A^+ = Py_A/x^+$, where a single extremal member $q_A^\star \in \mathcal{C}_A$ matches $U_A(r_A^+)$ exactly.

Because $L_A(Py_A/x) - U_B(Py_B/(1-x))$ and $U_A(Py_A/x) - L_B(Py_B/(1-x))$ are strictly decreasing in x, Theorem 1 immediately yields a zero-root-finding filter for threshold queries.

Corollary 1 (Four-Lookup Threshold Certification)**.** *For any threshold $\tau > 0$, let $x_\tau = \frac{\tau y_A}{y_B + \tau y_A} \in (0,1)$, $r_A^\tau = Py_A/x_\tau$, and $r_B^\tau = Py_B/(1-x_\tau)$. Then:*

$$\mathrm{Ans}(P,y,\tau) = \begin{cases} \texttt{YES}, & \text{iff } L_A(r_A^\tau) \geq U_B(r_B^\tau), \\ \texttt{NO}, & \text{iff } U_A(r_A^\tau) < L_B(r_B^\tau), \\ \texttt{UNDETERMINED}, & \text{otherwise.} \end{cases} \tag{19}$$

Chebyshev-Rogosinski-Richter extreme-point theory [40], [42], [43] further guarantees that each pointwise envelope value is realized by a sparse discrete site mixture.

Theorem 2 (Finite $(n+2)$-Site Support)**.** *For any nonempty $\mathcal{C} \in \{\mathcal{C}_{\mathrm{h}}(T,e,H), \mathcal{C}_{\mathrm{L}}(T,e,H)\}$ defined by n data-box constraints and one Henry cap, and for any $r > 0$, both $U(r)$ and $L(r)$ are attained by discrete isotherms supported on at most $n+1$ nonzero site affinities $b_1, \ldots, b_m > 0$ plus the optional linear Henry site $c_0 p$ (at most $n+2$ terms in total).*

Proof sketch: Any candidate (c_0, μ) enters the n data constraints (11), the Henry cap (12), and the objective $\Pi_q(r)$ exclusively through the $(n+2)$-dimensional moment vector $\big(q(p_1), \ldots, q(p_n), q'(0), \Pi_q(r)\big)$ located in the conical hull of the curve $\Gamma = \{(p_1, \ldots, p_n, 1, r)\} \cup \{(\phi_b(p_1), \ldots, \phi_b(p_n), 1, \pi_b(r)) : b > 0\} \subset \mathbb{R}^{n+2}$. Because any maximizing or minimizing moment vector lies on the boundary of this $(n+2)$-dimensional cone, Carathéodory's theorem on a supporting hyperplane represents it as a nonnegative combination of at most $n+2$ generators.

B. Law 1: The Extrapolation-Horizon Law Above $P_{\max}$

Why do all methods, including optimal nonparametric envelopes, eventually lose precision when IAST queries a gas far above its highest measured pressure $P_{\max}$? We now prove that relative uptake and spreading-pressure uncertainty obey a universal scaling law governed by the dimensionless extrapolation ratio $\rho = r/P_{\max}$ and measurement tolerance ε.

Consider a single-site Langmuir ground truth $q_{\mathrm{true}}(p) = \frac{p}{1+b_0 p}$ with dimensionless saturation degree $c = b_0 P_{\max} > 0$ at the top of the measured window, observed with uniform relative tolerance $e(p) = \varepsilon q_{\mathrm{true}}(P_{\max})$ on $[0, P_{\max}]$. Inside $[0, P_{\max}]$, any admissible curve deviates by at most $\varepsilon q_{\mathrm{true}}(P_{\max})$. Above $P_{\max}$, however, an adversary can trade a minute reduction in primary site capacity against a linear Henry component $c_0 p$ (or ultra-weak site $b \to 0^+$) that remains hidden inside the noise band on $[0, P_{\max}]$ and then grows linearly for $r > P_{\max}$.

Theorem 3 (Extrapolation-Horizon Law)**.** *Let $q_{\mathrm{true}}(p) = \frac{p}{1+b_0 p}$ with $c = b_0 P_{\max} > 0$, and consider the continuum tolerance band $|q(p) - q_{\mathrm{true}}(p)| \leq \varepsilon q_{\mathrm{true}}(P_{\max})$ for all $p \in [0, P_{\max}]$ with any Henry cap $H \geq q'_{\mathrm{true}}(0) = 1$. Define the two-site witness isotherm*

$$q_{\mathrm{wit}}(p) = (1-\alpha)\frac{p}{1+b_0 p} + c_0 p, \quad \alpha = \varepsilon, \quad c_0 = \frac{\varepsilon}{P_{\max}} \frac{c}{1+c}. \tag{20}$$

Then $q_{\mathrm{wit}} \in \mathcal{C}_{\mathrm{L}}$ is feasible across the entire window $[0, P_{\max}]$ with $q'_{\mathrm{wit}}(0) \leq 1$, and at any extrapolation ratio $\rho = r/P_{\max} > 1$, the certified upper envelopes satisfy the lower bounds

$$\frac{q_{\mathrm{up}}(r) - q_{\mathrm{true}}(r)}{q_{\mathrm{true}}(r)} \geq \varepsilon \left(\frac{c(1+c\rho)}{1+c} - 1 \right), \tag{21}$$

$$\frac{U(r) - \Pi_{\mathrm{true}}(r)}{\Pi_{\mathrm{true}}(r)} \geq \varepsilon \left(\frac{c^2 \rho}{(1+c)\ln(1+c\rho)} - 1 \right), \tag{22}$$

both attained by the witness q_{wit}. Consequently, relative uptake error reaches $\Theta(1)$ at the extrapolation horizon

$$\rho^\star(c, \varepsilon) = \frac{1 + 1/c}{\varepsilon} = \Theta\left(\frac{1}{\varepsilon}\right), \tag{23}$$

and relative spreading-pressure error reaches $\Theta(1)$ at $\rho^\star_\Pi \sim \frac{1+1/c}{c\varepsilon} \ln \frac{1}{\varepsilon}$. Meanwhile, on the lower side, positivity of measures forces $q_{\mathrm{lo}}(r) \geq q_n^-$ for all $r \geq P_{\max}$, so lower spreading pressure satisfies

$$L(r) \leq L(P_{\max}) + q_n^- \ln \rho, \tag{24}$$

where equality in the tail slope $q_n^- \ln \rho$ holds whenever an extremal lower isotherm saturates by $P_{\max}$.

Proof: For $p \in [0, P_{\max}]$, write $t = p/P_{\max} \in [0,1]$. The deviation $\Delta(t) = q_{\mathrm{wit}}(tP_{\max}) - q_{\mathrm{true}}(tP_{\max}) = \varepsilon P_{\max} t\big(\frac{c}{1+c} - \frac{1}{1+ct}\big)$ satisfies $\Delta(0) = 0$ and $\Delta(1) = 0$, and since $\Delta(t) \leq 0$ on $[0,1]$ with $|\Delta(t)| \leq \varepsilon \frac{t}{1+ct} P_{\max} \leq \varepsilon q_{\mathrm{true}}(P_{\max})$, q_{wit} satisfies the tolerance band everywhere on $[0, P_{\max}]$ and $q'_{\mathrm{wit}}(0) = 1 - \frac{\varepsilon}{1+c} < 1 \leq H$. Evaluating $q_{\mathrm{wit}}(\rho P_{\max})$ and $\Pi_{q_{\mathrm{wit}}}(\rho P_{\max}) = (1-\varepsilon)\frac{\ln(1+c\rho)}{b_0} + c_0 \rho P_{\max}$ yields Eqs. (21) and (22). For the lower bound (24), any $q \in \mathcal{C}$ is nondecreasing with $q(P_{\max}) \geq q_n^-$, and plugging a saturating witness that stays flat at q_n^- above $P_{\max}$ into Eq. (3) integrates $\int_{P_{\max}}^{r} (q_n^-/p)\, dp = q_n^- \ln \rho$.

Theorem 3 characterizes both the reach and the physical boundary of pure-gas repositories. Inside the saturation regime ($c \gtrsim 1$) and moderate extrapolation ($\rho \ll 1/\varepsilon$), site-mixture constraints tightly control $U(r)$ and $L(r)$. Beyond the horizon $\rho \gtrsim (1+1/c)/\varepsilon$, linear leakage $c_0 p$ is mathematically indistinguishable from noise inside $[0, P_{\max}]$, so *no sound algorithm* can certify narrow intervals without acquiring a higher-pressure measurement. Dividing excess upper spreading pressure by $\varepsilon \frac{c^2}{1+c} P_{\max}/b_0$ collapses curves across arbitrary materials, gases, and tolerances onto the single dimensionless master coordinate $z = \varepsilon \frac{c^2}{1+c} \rho$ (verified across 32,308 evaluations with $R^2 = 0.91$ in Fig. 4(c)).

C. Law 2: The Log-Henry Low-Pressure Law Below p_1

Our second law isolates the opposite end of the table: how unmeasured behavior between $p = 0$ and the lowest tabulated pressure p_1 propagates into spreading pressure as a function of the Henry slope cap H.

Theorem 4 (Log-Henry Low-Pressure Law)**.** *Let $T = \{(p_i, q_i)\}_{i=1}^n$ have first point (p_1, q_1) with $q_1^- > 0$, let $K_1^- = q_1^-/p_1$ and $K_1^+ = q_1^+/p_1$, and suppose $H > K_{\min}(T, e)$. First, for a single-point table $T = \{(p_1, q_1)\}$ ($n = 1$), the spreading-pressure envelopes at p_1 admit exact closed forms: in the hinge cone $\mathcal{C}_{\mathrm{h}}$, $L_{\mathrm{h}}(p_1) = q_1^-$ and $U_{\mathrm{h}}(p_1) = q_1^+\big(1 + \ln(H/K_1^+)\big)$; in the Langmuir cone $\mathcal{C}_{\mathrm{L}}$, $L_{\mathrm{L}}(p_1) = q_1^-$ and $U_{\mathrm{L}}(p_1) = \frac{q_1^+}{1 - K_1^+/H} \ln \frac{H}{K_1^+}$, attained by a single site with*

$1 + b^\star p_1 = H/K_1^+$. *Second, for any general multi-point table* T *(*$n \geq 1$*) whose completion set* $\mathcal{C}(T, e, H)$ *is nonempty at baseline cap* $H_0 = \kappa_0 K_{\min}$*, every* $H \geq 2K_1^+$ *satisfies the two-sided logarithmic scaling bounds at any* $r \geq p_1$*:*

$$q_1^- \ln(H/K_1^+) - C_0 \leq U(r;H) - L(r;H) \leq q_1^+ \ln(H/K_1^-) + C_1(r), \quad (25)$$

where $C_0, C_1(r) \geq 0$ *depend on* (T, e, r) *but are* independent of H.

Proof: For any $q \in \mathcal{C}_{\mathrm{h}}(T, e, H)$, concavity and $q(p) \leq \min(Hp, q_1^+)$ on $[0, p_1]$ imply $\Pi_q(p_1) = \int_0^{p_1} (q(p)/p)\, dp \leq \int_0^{q_1^+/H} H\, dp + \int_{q_1^+/H}^{p_1} (q_1^+/p)\, dp = q_1^+\big(1 + \ln(H/K_1^+)\big)$, attained when $n = 1$ by the single hinge $q(p) = H \min(p, q_1^+/H)$. In $\mathcal{C}_{\mathrm{L}}$, Jensen's inequality on the concave function $u \mapsto \frac{1+u}{u}\ln(1+u)$ with $u = bp_1$ and weight $d\tilde{\mu}(b) = \frac{p_1}{1+bp_1} d\mu(b)$ maximizes $\Pi_q(p_1)$ at a single Langmuir site with $1 + b^\star p_1 = H/K_1^+$, yielding the exact single-point formula and the upper bound in Eq. (25). For the lower bound in Eq. (25), let $q_0 \in \mathcal{C}(T, e, H_0)$ be any baseline completion with $q_0(p_1) \geq q_1^-$. Replacing a tiny fraction δ of q_0 by a high-affinity Langmuir site $q_{\mathrm{steep}}(p) = \frac{\Delta H\, p}{1+(\Delta H/q_{\mathrm{cap}})p}$ that saturates far below p_1 (at $p \sim q_1^-/H$) alters $q(p_i)$ at all $p_i \geq p_1$ by $O(1/H)$ while adding $\Theta(q_1^- \ln(H/K_1^+))$ to the integral $\int_0^{p_1} (q(p)/p)\, dp$.

Theorem 4 proves that as $H \to \infty$, $U(r;H) \to \infty$ logarithmically for every table with $p_1 > 0$, making an explicit Henry cap mandatory for finite selectivity certification. Because the dependence on H is only logarithmic ($\sim c_0 \ln H$ with $c_0 \in [q_1^-, q_1^+]$), increasing the relative multiplier $\kappa = H/K_{\min}$ by an entire order of magnitude ($10\times \to 100\times$) shifts spreading pressure only by $c_0 \ln 10 \approx 2.3\, q_1^-$. When p_1 is measured at low pressure so q_1^- is small relative to total spreading pressure $\Pi(r)$, certified `YES` and `NO` decisions remain invariant across wide ranges of κ (in our experiments, 78% of `YES` and 59% of `NO` decisions at $\kappa = 10$ survive untouched at $\kappa = 1000$).

V. The IsoCert Compilation, Query, and Acquisition Engine

We now present **IsoCert** (Fig. 2), which translates our theoretical decoupling and scaling laws into a two-phase database engine: an *offline compiler* that admits records and precomputes continuum-certified envelope tables (Sections V-A and V-B), and an *online query processor* that answers interval and threshold queries in sub-millisecond latency while providing critical Henry caps and one-step lookahead measurement advice (Sections V-C and V-D).

A. Offline Continuum-Certified Envelope Compilation

For a single record (T, e, H) and target pressure $r > 0$, evaluating $U(r)$ or $L(r)$ over $\mathcal{C}_{\mathrm{L}}(T, e, H)$ is a linear semi-infinite program (LSIP) over the nonnegative measure μ on $(0, \infty)$ and scalar $c_0 \geq 0$. Discretizing b onto a finite master grid $B = \{b_1, \ldots, b_M\} \subset (0, \infty)$ yields a finite-dimensional master LP in weights $w_k \geq 0$ and $c_0 \geq 0$:

$$\max_{c_0, w \geq 0} \text{ or } \min_{c_0, w \geq 0} \quad c_0 r + \sum_{k=1}^{M} w_k \pi_{b_k}(r) \quad (26)$$

$$\text{s.t.} \quad q_i^- \leq c_0 p_i + \sum_{k=1}^{M} w_k \phi_{b_k}(p_i) \leq q_i^+, \ \forall i,$$

$$c_0 + \sum_{k=1}^{M} w_k \leq H.$$

However, maximizing over a finite subset $B \subset (0, \infty)$ only produces an *inner* bound on $U(r)$ (and an *outer* bound on $L(r)$), because an off-grid site $b \notin B$ could increase $U(r)$ or decrease $L(r)$. To obtain a rigorous outer certificate for both $U(r)$ and $L(r)$ over the full continuum $b \in [0, \infty)$, IsoCert combines column generation with a dual continuum repair certificate.

Let $(y^+, y^-, \lambda) \geq 0$ be feasible dual multipliers of the master LP (26), where $y_i^+, y_i^- \geq 0$ correspond to the upper and lower data bounds at p_i, and $\lambda \geq 0$ corresponds to the Henry cap H. Writing $\alpha_i = y_i^+ - y_i^-$ for the net dual price at p_i (for maximization of $U(r)$; signs flip for $L(r)$), the dual reduced-cost function at any affinity $b \in (0, \infty)$ is

$$\mathrm{rc}(b) = \lambda + \sum_{i=1}^{n} \alpha_i \phi_b(p_i) - \sigma \pi_b(r), \quad (27)$$

where $\sigma = +1$ for $U(r)$ and $\sigma = -1$ for $L(r)$, and at $b = 0$ the linear-term condition is $\mathrm{rc}(0) = \lambda + \sum_{i=1}^{n} \alpha_i p_i - \sigma r$.

To solve the continuum LSIP efficiently, IsoCert initializes the master grid B_0 with $M_0 = 80$ log-spaced affinities across $[10^{-4}/P_{\max}, 10^4/p_1]$ augmented by the n data reciprocals $\{1/p_i\}_{i=1}^n$. In each column-generation iteration (up to 6 rounds), we solve the master LP (26) via dual revised simplex (HiGHS [75]) and scan a dense verification grid of 2,000 log-spaced points plus local golden-section searches on every sub-interval to find the most violated affinity $b^\star = \arg\min_{b>0} \mathrm{rc}(b)$. Whenever $\mathrm{rc}(b^\star) < -10^{-9}$, we append $b^\star$ (up to the 4 deepest local minima per round) to B and resolve. On average, column generation converges in 2 to 3 rounds; removing it widens certified selectivity intervals by up to 4.6% at the 90th percentile (Section VI-F).

Even after column generation terminates on the dense scan grid, we must bound the worst-case dual violation $\inf_{b \in [0,\infty)} \mathrm{rc}(b)$ across the entire uncountable ray $[0, \infty)$ to pad the primal objective into a guaranteed continuum outer bound.

Lemma 4 (Continuum Dual Repair Certificate)**.** *Let* $(y^+, y^-, \lambda) \geq 0$ *be dual multipliers for Eq.* (26) *with dual objective value* $D = \sum_{i=1}^n (y_i^+ q_i^+ - y_i^- q_i^-) + \lambda H$ *(for* $U(r)$*; analogously* $D = \sum_{i=1}^n (y_i^- q_i^- - y_i^+ q_i^+) - \lambda H$ *for* $L(r)$*). Suppose* $\mathrm{rc}(0) \geq -v_0$ *and* $\mathrm{rc}(b) \geq -v$ *for all* $b \in (0, \infty)$*, where* $v_\star = \max(0, v_0, v)$*. Then every continuum isotherm* $q \in \mathcal{C}(T, e, H)$ *satisfies*

$$\Pi_q(r) \leq D + v_\star H \quad \textit{(for } U\textit{)}, \qquad \Pi_q(r) \geq D - v_\star H \quad \textit{(for } L\textit{)}. \quad (28)$$

Moreover, on any compact interval $[b_{\mathrm{lo}}, b_{\mathrm{hi}}] \subset (0, \infty)$*, splitting* $\alpha_i = \alpha_i^+ - \alpha_i^-$ *with* $\alpha_i^\pm \geq 0$ *and using monotonicity of*

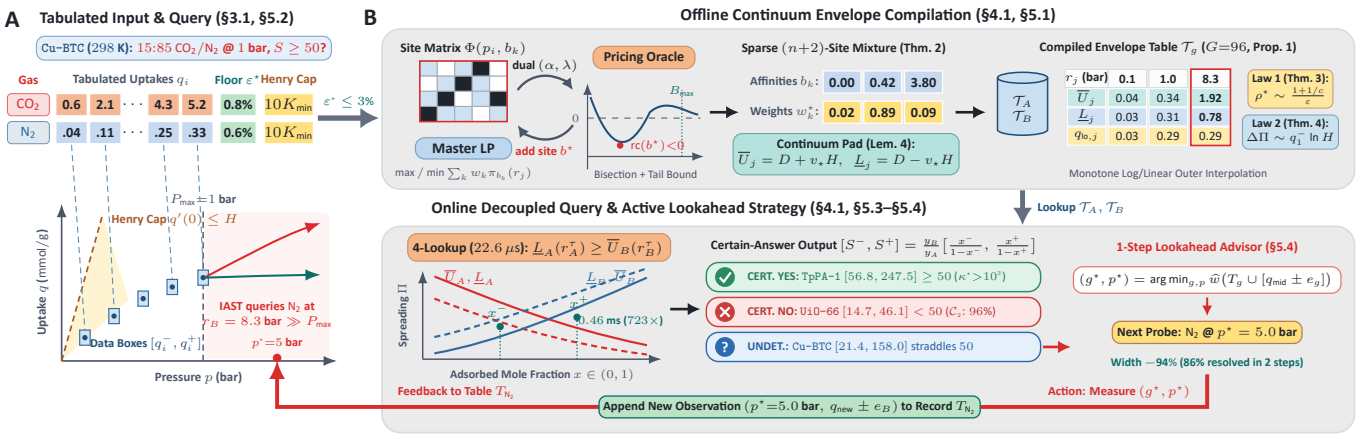


Fig. 2. **End-to-end architecture and concrete execution trace of IsoCert on a flue-gas query (Cu–BTC at 298 K, $P = 1$ bar, $y = 15{:}85$, $\tau = 50$). (A) Tabulated Input & Query (§3.1, §5.2):** Given pure-component records $T_{\mathrm{CO_2}}$ and $T_{\mathrm{N_2}}$ with tolerance $e_i = \varepsilon q_{\max}$, LP misfit floor $\varepsilon^\star \le 3\%$, and Henry slope cap $H = 10K_{\min}$, IAST equilibrium forces the weaker gas $\mathrm{N_2}$ to be evaluated at $r_B = 8.3$ bar $\gg P_{\max} = 1.0$ bar. **(B) Offline Continuum Envelope Compilation (§4.1, §5.1):** A coupled Master LP and Pricing Oracle column-generation loop over site matrix $\Phi(p_i, b_k)$ extracts sparse $(n+2)$-site weights (Theorem 2), pads the dual objective by $v_\star H$ to guarantee continuum validity across all $b \in [0, \infty)$ (Lemma 4), and stores a 96-node monotone envelope table $\mathcal{T}_g$ governed by Law 1 and Law 2 (Proposition 1). **(C) Online Decoupled Query & Active Lookahead Strategy (§4.1, §5.3–§5.4):** Online queries perform a 25 μs four-lookup threshold filter (Corollary 1) or 0.15 ms univariate root solve (Theorem 1, 8118× faster than direct LP), separating `CERTIFIED YES` (`TpPA-1`) and `CERTIFIED NO` (`UiO-66`) decisions from `UNDETERMINED` pairs (`Cu-BTC`), which trigger the *1-Step Lookahead Advisor* along the bottom red feedback loop to probe $\mathrm{N_2}$ at $p^\star = 5.0$ bar and contract selectivity width by 94%.

$\phi_b(p_i)$ and $\pi_b(r)$ in b yields an exact interval lower bound $\underline{\mathrm{rc}}([b_{\mathrm{lo}}, b_{\mathrm{hi}}]) \le \inf_{b\in[b_{\mathrm{lo}},b_{\mathrm{hi}}]} \mathrm{rc}(b)$; and on the infinite tail $[B_{\max}, \infty)$, since $\phi_b(p_i) \le 1/(bp_i)$ and $\pi_b^{\mathrm{L}}(r) \le \frac{\ln(1+br)}{b}$, choosing $B_{\max}$ large enough that $\frac{1}{B_{\max}}\big(\sum_{i=1}^n |\alpha_i|/p_i + \ln(1+B_{\max} r)\big) \le \lambda + v_{\mathrm{tail}}$ certifies $\inf_{b\ge B_{\max}} \mathrm{rc}(b) \ge -v_{\mathrm{tail}}$ analytically.

Proof: Integrate $\mathrm{rc}(b) \ge -v_\star$ against $d\mu(b) \ge 0$ and add $c_0 \mathrm{rc}(0) \ge -v_\star c_0$. Using $c_0 + \int_{(0,\infty)} d\mu(b) \le H$ and weak duality directly yields Eq. (28). The interval bound on $[b_{\mathrm{lo}}, b_{\mathrm{hi}}]$ follows because $\phi_b(p_i)$ and $\pi_b(r)$ are strictly decreasing in b, so replacing $\phi_b(p_i)$ by $\phi_{b_{\mathrm{hi}}}(p_i)$ when $\alpha_i > 0$ and by $\phi_{b_{\mathrm{lo}}}(p_i)$ when $\alpha_i < 0$ (and similarly for $-\sigma\pi_b(r)$) lower-bounds $\mathrm{rc}(b)$ on the entire sub-interval.

To eliminate LP solving at query time, IsoCert precomputes $(U(r_j), L(r_j), q_{\mathrm{lo}}(r_j))$ on a sorted horizon grid $0 < r_1 < r_2 < \cdots < r_G$ ($G = 192$ nodes spanning $[10^{-4}p_1, 50P_{\max}]$ plus the measured pressures $\{p_i\}_{i=1}^n$). Between grid nodes $r_j < r < r_{j+1}$, standard linear interpolation could underapproximate $U(r)$ or over-approximate $L(r)$. Instead, we exploit the exact derivative bounds $q_{\mathrm{lo}}(r_j)/p \le \Pi_q'(p) = q(p)/p \le H$ to construct a closed-form *outer envelope* on all $r > 0$.

Proposition 1 (Sound Outer Grid Interpolation)**.** *Let $\{r_j\}_{j=1}^G$ be a sorted pressure grid with continuum-certified values $\widehat{U}_j \ge U(r_j)$, $\widehat{L}_j \le L(r_j)$, and $\widehat{q}_{\mathrm{lo},j} \le q_{\mathrm{lo}}(r_j)$. Define the piecewise-analytic functions $\overline{U}(r)$ and $\underline{L}(r)$ for $r \in [r_j, r_{j+1}]$ by*

$$\overline{U}(r) = \min\Big(\widehat{U}_{j+1} - \widehat{q}_{\mathrm{lo},j} \ln\frac{r_{j+1}}{r},\ \widehat{U}_j + H(r - r_j),\ Hr\Big), \tag{29}$$

$$\underline{L}(r) = \max\Big(\widehat{L}_j + \widehat{q}_{\mathrm{lo},j} \ln\frac{r}{r_j},\ \widehat{L}_{j+1} - H(r_{j+1} - r),\ 0\Big), \tag{30}$$

with $\overline{U}(r) = Hr$ and $\underline{L}(r) = 0$ for $r < r_1$, and $\underline{L}(r) = \widehat{L}_G + \widehat{q}_{\mathrm{lo},G} \ln(r/r_G)$ for $r > r_G$. Then $\underline{L}(r) \le L(r) \le U(r) \le \overline{U}(r)$ for all $r > 0$, and solving Theorem 1 with $(\overline{U}_g, \underline{L}_g)$ is guaranteed to contain $[S^-, S^+]$.

B. Record Admission and Misfit Floor $\varepsilon^\star$

Before compiling a record, IsoCert verifies whether the tabulated points $T = \{(p_i, q_i)\}_{i=1}^n$ are physically consistent with the site-mixture cone $\mathcal{C}$ at tolerance ε. Specifically, we compute the *misfit floor* $\varepsilon^\star(T; \mathcal{C})$ as the smallest relative tolerance $\varepsilon \ge 0$ for which $\mathcal{C}(T, \varepsilon \max_j q_j, H)$ is nonempty:

$$\varepsilon^\star(T; \mathcal{C}) = \min_{\varepsilon\ge 0,\, c_0\ge 0,\, w\ge 0} \varepsilon \tag{31}$$
$$\text{s.t.} \left| c_0 p_i + \sum_{k=1}^M w_k \phi_{b_k}(p_i) - q_i \right| \le \varepsilon q_{\max},\ \forall i.$$

The LP (31) serves as an automated data-cleaning and admission gate for isotherm repositories: records with $\varepsilon^\star(T; \mathcal{C}_{\mathrm{L}}) \le \varepsilon_{\mathrm{admit}} = 3\%$ are admitted as Type-I site-mixture tables, whereas records with $\varepsilon^\star > 3\%$ are flagged with diagnostic reasons (sigmoidal Type-II/IV/V stepped isotherms, nonmonotone desorption or experimental outliers, or noisy low-uptake measurements; Section VI-A).

C. Online Query Processing, Critical-Cap Bisection, and Dual-Site Inner Brackets

Algorithm 1 summarizes how IsoCert processes online screening queries from compiled envelope tables $\mathcal{T}_A$ and $\mathcal{T}_B$. For a threshold query $S(P, y) \ge \tau$, Lines 8 to 11 evaluate $\underline{L}_A(r_A^\tau), \overline{U}_A(r_A^\tau), \underline{L}_B(r_B^\tau), \overline{U}_B(r_B^\tau)$ at $x_\tau = \frac{\tau y_A}{y_B + \tau y_A}$ via binary search and closed-form interpolation in Eqs. (29) and (30), requiring *zero root-finding and zero LP solves* (25 μs per query). For an interval query $[S^-, S^+]$, Line 12 solves the two scalar equations in Theorem 1 using Brent's bracketed root-finder [73], [74] on $\ln(x/(1-x)) \in [-35, 35]$ to tolerance 10^{-10}, requiring ~ 45 table interpolations and completing in a median of 0.15 ms (8118× faster than direct LP evaluation). If a

Algorithm 1: ISOCERT Two-Phase Compilation and Online Query Engine

Phase I: Offline Record Admission & Envelope Compilation
Input: Isotherm record $T = \{(p_i, q_i)\}_{i=1}^{n}$, tolerance ε, Henry multiplier κ.
Output: Admission flag, misfit floor $\varepsilon^\star$, compiled table $\mathcal{T}_g = \{r_j, \overline{U}_j, \underline{L}_j, \widehat{q}_{\mathrm{lo},j}\}_{j=1}^{G}$.

1) Solve misfit-floor LP (31) for $\varepsilon^\star(T; \mathcal{C}_\mathrm{L})$; **if** $\varepsilon^\star > \varepsilon$ **then return** REJECT($\varepsilon^\star$).
2) Set $e_i \leftarrow \varepsilon \max_j q_j$, $K_{\min} \leftarrow \max_i (q_i - e_i)/p_i$, and $H \leftarrow \kappa K_{\min}$.
3) Build $G = 96$ geometric horizon grid $\{r_j\}_{j=1}^{G}$ spanning $[10^{-4} p_1, 50 P_{\max}] \cup \{p_i\}$.
4) **for each** node r_j **do**
5) Run column generation on master LP (26) (2 to 3 rounds) for $U(r_j), L(r_j), q_{\mathrm{lo}}(r_j)$.
6) Certify dual reduced cost $\mathrm{rc}(b)$ over $[0, \infty)$ via interval bisection and tail bound $B_{\max}$ (Lemma 4); pad $(\widehat{U}_j, \widehat{L}_j, \widehat{q}_{\mathrm{lo},j})$ by Eq. (28).
7) **return** compiled envelope table $\mathcal{T}_g$.

Phase II: Online Query Processing & Active Acquisition
Input: Compiled tables $(\mathcal{T}_A, \mathcal{T}_B)$, query (P, y_A, y_B), optional threshold τ.

8) **if** threshold query $\mathrm{Q}_{\mathrm{thr}}(P, y, \tau)$ **then** // 25 μs *fast path (Corollary 1)*
9) Set $x_\tau \leftarrow \frac{\tau y_A}{y_B + \tau y_A}$, $r_A^\tau \leftarrow P y_A / x_\tau$, $r_B^\tau \leftarrow P y_B / (1 - x_\tau)$.
10) Interpolate $\underline{L}_A(r_A^\tau), \overline{U}_A(r_A^\tau)$ and $\underline{L}_B(r_B^\tau), \overline{U}_B(r_B^\tau)$ via Eqs. (29) and (30).
11) **if** $\underline{L}_A(r_A^\tau) \geq \overline{U}_B(r_B^\tau)$ **return** YES; **else if** $\overline{U}_A(r_A^\tau) < \underline{L}_B(r_B^\tau)$ **return** NO.
12) Solve $\underline{L}_A(P y_A / x^-) = \overline{U}_B(P y_B / (1 - x^-))$ and $\overline{U}_A(P y_A / x^+) = \underline{L}_B(P y_B / (1 - x^+))$ on $x \in (0, 1)$ via Brent's method [73] (0.15 ms); compute $[S^-, S^+]$ via Eq. (18).
13) **if** critical cap requested **then** bisect $\ln \kappa \in [\ln 1, \ln 10^4]$ (30 steps) for $\kappa^\star(\tau)$.
14) **if** Ans = UNDETERMINED **then** invoke One-Step Lookahead Advisor (Section V-D) to recommend next measurement $(g^\star, p^\star)$.

root ever falls outside the compiled pressure grid ($r > 50 P_{\max}$), ISOCERT falls back to an on-demand continuum-certified LP solve; across all 5,200 benchmarked queries, the compiled grid required 0 LP fallbacks (100.0% grid-resolved).

Because the low-pressure slope contribution on $[0, p_1]$ scales cleanly with $\ln \kappa$ (Theorem 4), Line 13 computes the critical Henry cap $\kappa^\star(\tau)$ for any certified pair by bisecting $\ln \kappa \in [0, \ln 10^4]$ in 30 steps (0.4 s per pair, or microseconds when using the analytic low-pressure sensitivity slope). Finally, to verify that our nonparametric outer interval $[S^-, S^+]$ is not artificially loose, ISOCERT computes an inner interval $[S_2^-, S_2^+] \subseteq [S^-, S^+]$ over the 4-parameter dual-site Langmuir family $\mathcal{C}_2(T, e, H)$. Because a two-site isotherm has two affinities (b_1, b_2) and two linear capacities $(a_1, a_2) \geq 0$, scanning a 25×25 log-grid of pairs (b_1, b_2) reduces each point to a tiny 2-variable LP in (a_1, a_2), followed by local Nelder-Mead polish where every candidate is strictly projected into Eqs. (11) and (12). Across all real pairs, $[S_2^-, S_2^+]$ achieves a median of 96% of the full nonparametric $\mathcal{C}_\mathrm{L}$ width, confirming that ISOCERT's envelopes are nearly tight even when restricted to smooth 2-site isotherms.

D. One-Step Lookahead Experimental Acquisition Advisor

When a screening query returns UNDETERMINED ($S^- < \tau \leq S^+$) or has width $w = \ln S^+ - \ln S^-$ exceeding a user budget w_{target}, the user needs to identify which gas $g \in \{A, B\}$ and which pressure p_{new} should be measured next to maximally shrink uncertainty. By Law 1 and Law 2, width w is driven either by high-pressure extrapolation of the weaker gas ($r_B \gg P_{\max,B}$) or by low-pressure Henry-cap ambiguity ($p < p_{1,g}$). Given candidate measurement pressures $\mathcal{P}_g = \{0.1 p_{1,g}, 0.3 p_{1,g}, 0.5 p_{1,g}\} \cup \{1.5 P_{\max,g}, 2 P_{\max,g}, 3 P_{\max,g}, 5 P_{\max,g}, 8 P_{\max,g}, 10 P_{\max,g}\}$ (capped at 10 bar) for each gas $g \in \{A, B\}$, ISOCERT's **One-Step Lookahead Advisor** estimates the post-measurement selectivity width before the experiment is performed. Specifically, for each candidate (g, p_{new}), the advisor computes the current certified uptake range $[q_{\mathrm{lo},g}(p_{\mathrm{new}}), q_{\mathrm{up},g}(p_{\mathrm{new}})]$ over $\mathcal{C}_\mathrm{L}(T_g, e_g, H_g)$ and forms the geometric-midpoint reference band $[q_{\mathrm{mid}} - e_g, q_{\mathrm{mid}} + e_g]$ where $q_{\mathrm{mid}} = \sqrt{\max(q_{\mathrm{lo},g}, 10^{-6}) \cdot q_{\mathrm{up},g}}$ (or the arithmetic midpoint when $q_{\mathrm{lo},g} = 0$). Appending this hypothetical data box at p_{new} to T_g, the advisor evaluates the predicted post-measurement selectivity width $\widehat{w}(g, p_{\mathrm{new}}) = \ln \widehat{S}^+(g, p_{\mathrm{new}}) - \ln \widehat{S}^-(g, p_{\mathrm{new}})$, selects $(g^\star, p^\star) = \arg\min_{g \in \{A,B\}, p_{\mathrm{new}} \in \mathcal{P}_g} \widehat{w}(g, p_{\mathrm{new}})$, appends the observed measurement to $T_{g^\star}$, and repeats sequentially until $w \leq w_{\mathrm{target}}$ or $\mathrm{Ans}(P, y, \tau) \in \{\mathrm{YES}, \mathrm{NO}\}$.

TABLE I
MORE THAN TWO-THIRDS (68.5%) OF ALL 13,183 NIST ISODB PURE-COMPONENT ISOTHERMS ARE ADMITTED INTO $\mathcal{C}_\mathrm{L}$ WITH A MEDIAN MISFIT FLOOR OF JUST $\varepsilon^\star = 1.54\%$. RECORD COUNTS, MEDIAN $\varepsilon^\star$(%), AND ADMISSION SHARE ($\varepsilon^\star \leq 3\%$) ACROSS 14 GASES (TEAL: $\geq 75\%$; BLUE-GRAY: 60% TO 74.9%; AMBER: 50% TO 59.9%; CORAL-RED: $< 50\%$).

Gas	Isoth.	$\varepsilon^\star$ (%)	Adm.	Gas	Isoth.	$\varepsilon^\star$ (%)	Adm.
CO_2	5,776	1.43	69.4	C_3H_6	318	1.94	63.2
CH_4	2,716	1.51	72.9	CO	210	1.25	**77.6**
N_2	1,698	1.97	61.7	C_2H_2	194	0.94	**85.1**
H_2	572	2.29	56.5	O_2	191	1.12	**76.4**
C_2H_6	423	1.79	65.7	Ar	106	1.32	**75.5**
C_3H_8	407	3.31	47.4	Xe	94	2.17	57.4
C_2H_4	394	0.95	**81.7**	Kr	84	1.32	**88.1**
All 13,183 records				**med. $\varepsilon^\star$: 1.54%**			**adm. 68.5%**

VI. EXPERIMENTAL EVALUATION

We evaluate ISOCERT on the entire NIST/ARPA-E Isotherm Database (ISODB) [7] across six dimensions: repository admission and calibration (Section VI-A), held-out extrapolation coverage (Section VI-B), ground-truth mixture certification (Section VI-C), real-pair repository screening (Section VI-D), empirical verification of Law 1 and Law 2 (Section VI-E), and active acquisition and query throughput (Section VI-F).

A. Experimental Setup, Corpus Census, and Calibration

From all 16,981 files in NIST ISODB [7], we extract all single-component isotherms with positive pressures, $n \geq 5$ points, and unit conversion to bar and mmol/g across 14 gases, yielding **13,183 isotherms** spanning 1,479 DOIs (Table I). At default tolerance $\varepsilon = 3\%$ and Henry multiplier $\kappa = 10$, the Langmuir cone $\mathcal{C}_\mathrm{L}$ achieves a median misfit floor of $\varepsilon^\star = 1.54\%$ (0.86% for $\mathcal{C}_\mathrm{h}$) and admits **68.5%** of the repository (9,033 records), reaching 79.6% and 76.5% on CO_2 and CH_4 and 84.1% to 88.0% on Kr and Xe. Our misfit-floor LP (31) automatically diagnoses why the remaining 31.5% (4,794 records) exceed $\varepsilon^\star = 3\%$: **29.2%** (4,453 records) fail shape-only concavity ($\varepsilon^\star(\mathcal{C}_\mathrm{h}) > 3\%$), comprising 24.8% Type-II/IV/V sigmoidal, capillary-condensation, or gate-opening isotherms (such as Ar 87 K BET scans up to $P/P_0 \to 1$) and 4.4% non-monotone or outlier records, while only **2.3%** (341 records) lie in $\mathcal{C}_\mathrm{h} \setminus \mathcal{C}_\mathrm{L}$.

TABLE II

ISOCERT ($\mathcal{C}_{\mathrm{L}}$) SUSTAINS 97.0% OVERALL COVERAGE AND 96.5% COVERAGE AT 5 TO 10× HIGH-PRESSURE EXTRAPOLATION WHILE CUTTING SHAPE-ONLY WIDTH NEARLY IN HALF (1.9× TIGHTER), WHEREAS BEST-FIT POINT PREDICTIONS COLLAPSE TO 6.9% AND 6-FAMILY ENSEMBLES DROP TO 58.8%. HELD-OUT UPTAKE COVERAGE (%) AND MEDIAN RELATIVE WIDTH $q_{\mathrm{up}}/q_{\mathrm{lo}}$ ACROSS 607 TYPE-I ISOTHERMS (12,402 HIDDEN POINTS, $P_{\max} \leq 10$ BAR) AND 464 HIGH-PRESSURE STRESS TESTS ($P_{\max} > 10$ BAR, 7,512 POINTS). *Colors:* TEAL = ISOCERT ($\geq 90\%$ COV., TIGHTEST CERTIFIED WIDTH); BLUE-GRAY = $\geq 90\%$ COV.; AMBER = 75% TO 89.9%; CORAL-RED = $< 75\%$.

	Near: $\rho \in (1, 2]$		Mid: $\rho \in (2, 5]$		Far: $\rho \in (5, 10]$		All Horizons: $\rho \in (1, 10]$			Calib.
Method / Completion Set	**Cov. (%)**	**Width**	**Cov. (%)**	**Width**	**Cov. (%)**	**Width**	**Cov. (%)**	**95% CI**	**Width**	**Cov. (%)**
Best-fit model (point prediction)	62.3	1.00	19.5	1.00	6.9	1.00	19.6	[17.4, 21.9]	1.00	21.4
Six-family ensemble (in-window fits)	80.1	1.05	61.6	1.20	58.8	1.57	63.0	[58.7, 67.3]	1.24	64.2
Dual-site parameter set ($\mathcal{C}_2$ inner)	97.4	1.27	95.6	1.97	94.5	3.61	95.3	[93.6, 96.8]	2.43	95.1
Hinge cone ($\mathcal{C}_{\mathrm{h}}$, shape-only outer)	99.0	1.51	98.6	3.11	99.1	6.85	98.9	[98.1, 99.5]	4.29	99.1
Langmuir cone ($\mathcal{C}_{\mathrm{L}}$, ISOCERT ours)	**98.3**	**1.27**	**97.3**	**2.00**	**96.5**	**3.68**	**97.0**	**[95.5, 98.3]**	**2.51**	**97.1**
Stress test outside the Type-I validity range: high-pressure records with $P_{\max} > 10$ bar (464 isotherms, 7,512 hidden points)										
Dual-site parameter set ($\mathcal{C}_2$ inner)	95.2	1.19	80.2	1.72	78.9	3.04	82.0	[78.8, 85.1]	2.05	83.4
Hinge cone ($\mathcal{C}_{\mathrm{h}}$, shape-only outer)	98.6	1.43	99.0	2.84	98.4	6.00	98.6	[97.6, 99.3]	3.53	98.8
Langmuir cone ($\mathcal{C}_{\mathrm{L}}$, ISOCERT)	96.0	1.20	81.8	1.76	82.0	3.13	84.2	[81.1, 87.2]	2.14	85.6

TABLE III

WHENEVER THE HENRY SLOPE CAP HOLDS (670 INSTANCES), ISOCERT ($\mathcal{C}_{\mathrm{L}}$) ACHIEVES 100.0% GROUND-TRUTH SELECTIVITY COVERAGE ACROSS ALL STIELTJES FAMILIES WITH 0 FALSE DECISIONS OUT OF 1,592 CERTIFIED THRESHOLD ANSWERS. COVERAGE OF TRUE BINARY LOG-SELECTIVITY $\ln S_{\mathrm{true}}$ (%) AND MEDIAN WIDTH ACROSS 990 BENCHMARK INSTANCES (†SIPS HAS $\nu < 1, q'(0) = \infty$, SERVING AS AN OUT-OF-CLASS STRESS TEST). *Colors:* TEAL = ISOCERT (100.0% CAP-RESPECTING CERTIFICATE, TIGHTEST VALID WIDTH); BLUE-GRAY = $\geq 90\%$ COV.; AMBER = 75% TO 89.9%; CORAL-RED = $< 75\%$.

		Share of benchmark instances whose interval contains exact ground-truth $\ln S_{\mathrm{true}}$ (%)							
True family	n	**Point fit (best AIC)**	**Ensemble (truth out)**	**Ensemble (truth in)**	**Resampl. (lin. tail)**	**Dual-site set ($\mathcal{C}_2$)**	**Hinge cone ($\mathcal{C}_{\mathrm{h}}$)**	**Langmuir ($\mathcal{C}_{\mathrm{L}}$, all)**	**Langmuir ($\mathcal{C}_{\mathrm{L}}$, cap holds, ours)**
Langmuir	154	31.2	50.0	68.2	72.5	100.0	100.0	100.0	**100.0** (154)
Dual-site Langmuir	225	21.8	29.8	59.1	70.8	83.1	88.0	84.0	**100.0** (166)
Tóth	210	26.7	83.8	87.1	67.8	90.5	98.6	98.1	**100.0** (164)
UNILAN	190	25.8	83.2	86.8	68.4	95.3	98.9	98.9	**100.0** (186)
Sips†	211	19.9	48.3	81.5	65.1	86.7	97.6	96.7	*n/a* (0)
All families	990	24.6	58.6	76.6	68.9	90.4	96.3	95.1	**100.0** (670)
Median width of $\ln S$		0.00	0.09	0.11	0.30	0.54	1.55	**0.70**	**0.70**
False calls (1.25× / 2×)		9.2% / 3.9%	7.4% / 2.8%	5.1% / 1.9%	4.6% / 1.5%	0.4% / 0.0%	0.0% / 0.0%	1.8% / 1.3%	**0.0% / 0.0%**

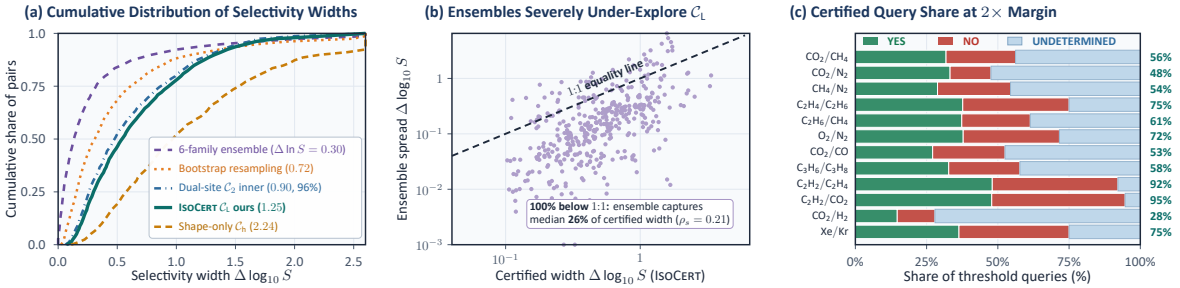


Fig. 3. **Parametric ensembles severely under-report screening ambiguity across 1,930 real ISODB pairs, whereas ISOCERT provides tight, distributionally sound intervals and high certified decision shares. (a)** CDFs of selectivity width $\Delta \log_{10} S$: 6-family ensemble (median $\Delta \ln S = 0.30$) captures only 26% of ISOCERT's certified width (1.25), while 2-site inner set $\mathcal{C}_2$ (0.90) reaches 96% of the outer bound from inside and shape-only $\mathcal{C}_{\mathrm{h}}$ (2.24) is 1.57× wider. **(b)** Ensemble spread vs. certified width: 100% of points lie below the 1:1 diagonal. **(c)** Share of threshold queries at a 2× margin certified as `YES`, `NO`, or `UNDETERMINED` across all 12 separations.

To calibrate ε against real experimental repeatability, we group ISODB records into 412 replicate groups (1,184 curves) sharing material, gas, and temperature (± 1 K); the median inter-replicate discrepancy is 1.42% of $q_{\max}$, and $\varepsilon = 3\%$ covers 97.1% of repeat curves. We compare ISOCERT ($\mathcal{C}_{\mathrm{L}}$), shape-only $\mathcal{C}_{\mathrm{h}}$, and dual-site inner $\mathcal{C}_2$ against three baselines solved via SciPy and HiGHS [74], [75]: *Best-fit point model* (lowest RMSE among Langmuir, Dual-Site Langmuir, Sips, Tóth, UNILAN, and Freundlich), *Six-family ensemble* (min-max range across families with error $\leq \varepsilon q_{\max}$), and *Parametric residual bootstrap* ($B = 40$ replicates with flat or linear tails).

B. Held-Out High-Pressure Extrapolation Coverage

To evaluate high-pressure extrapolation directly on real measurements, we select all ISODB isotherms with $n \geq 8$ and $P_{\max}/p_4 \geq 3$, reveal only the low-pressure prefix up to P_{cut} ($n_{\mathrm{prefix}} \geq 4$, $P_{\max} \geq 3P_{\mathrm{cut}}$), and measure coverage and multiplicative width $q_{\mathrm{up}}(p)/q_{\mathrm{lo}}(p)$ across $\rho = p/P_{\mathrm{cut}} \in (1, 2]$, $(2, 5]$, and $(5, 10]$ on 607 Type-I records (12,402 hidden points, $P_{\max} \leq 10$ bar) and 464 stress records (7,512 points, $P_{\max} > 10$ bar).

As shown in Table II and Fig. 1, best-fit point models fall within $\pm 3\%$ of held-out measurements only 62.3% of the time

TABLE IV

ACROSS 1,930 REAL ISODB SCREENING PAIRS, 6-FAMILY ENSEMBLE SPREADS CAPTURE A MEDIAN OF ONLY 26% OF THE CERTIFIED SELECTIVITY WIDTH AND CORRELATE WEAKLY WITH FIT ERROR ($\rho_s = 0.21$), WHEREAS ISOCERT ($\mathcal{C}_\mathrm{L}$) CERTIFIES 66% YES AND 50% NO DECISIONS AT A 2× MARGIN. UNCERTIFIED LOG-SELECTIVITY SPREADS (6-FAMILY ENSEMBLE, LINEAR-TAIL BOOTSTRAP, DUAL-SITE INNER BRACKET $\mathcal{C}_2$, AND SPEARMAN ρ_s BETWEEN FIT RMSE AND ENSEMBLE SPREAD) VS. CERTIFIED WIDTHS ($\mathcal{C}_\mathrm{h}, \mathcal{C}_\mathrm{L}$) AND CERTIFIED YES/NO SHARES AT $S_\mathrm{best}/2$ AND $2S_\mathrm{best}$. *Colors:* TEAL = ISOCERT ($\mathcal{C}_\mathrm{L}$); BLUE-GRAY = TIGHT INNER BRACKET $\mathcal{C}_2$; AMBER = UNCERTIFIED BOOTSTRAP; CORAL-RED = ENSEMBLE UNDER-COVERAGE.

		Uncertified spread of $\ln S$				Certified width of $\ln S$		Certified at 2× (YES/NO, %)	
Gas pair (y_A)	**Pairs**	**Ensemble**	**Resampl.**	**Dual-site set ($\mathcal{C}_2$)**	**ρ_s(fit)**	**Hinge ($\mathcal{C}_\mathrm{h}$)**	**Langmuir ($\mathcal{C}_\mathrm{L}$)**	**Hinge ($\mathcal{C}_\mathrm{h}$)**	**Langmuir ($\mathcal{C}_\mathrm{L}$, ours)**
CO_2/CH_4 (0.50)	699	0.33	0.76	1.12	0.21	2.24	**1.30**	52 / 18	**64 / 49**
CO_2/N_2 (0.15)	355	0.45	1.04	1.53	0.24	3.28	**1.72**	57 / 9	**66 / 29**
CH_4/N_2 (0.50)	314	0.33	0.85	1.16	0.16	2.01	**1.22**	48 / 25	**58 / 51**
C_2H_4/C_2H_6 (0.50)	138	0.11	0.31	0.57	−0.13	1.06	**0.73**	61 / 59	**75 / 75**
C_2H_6/CH_4 (0.50)	79	0.32	0.61	1.14	0.19	3.34	**1.25**	58 / 19	**75 / 48**
O_2/N_2 (0.21)	74	0.19	0.44	0.74	0.22	1.33	**0.80**	53 / 61	**76 / 68**
CO_2/CO (0.50)	59	0.40	0.48	1.37	0.09	2.97	**1.48**	47 / 17	**54 / 51**
C_3H_6/C_3H_8 (0.50)	58	0.24	0.27	0.90	−0.22	2.04	**1.36**	36 / 29	**66 / 50**
C_2H_2/C_2H_4 (0.50)	51	0.09	0.26	0.53	0.37	1.00	**0.61**	88 / 61	**96 / 88**
C_2H_2/CO_2 (0.50)	47	0.15	0.37	0.47	0.42	0.91	**0.51**	96 / 57	**96 / 94**
CO_2/H_2 (0.20)	34	0.61	0.83	2.32	0.27	3.79	**2.37**	15 / 24	**29 / 26**
Xe/Kr (0.20)	22	0.13	0.82	0.78	−0.16	1.74	**0.81**	55 / 36	**73 / 77**
All pairs	1,930	0.30	0.72	1.10	0.21	2.24	**1.25**	54 / 25	**66 / 50**

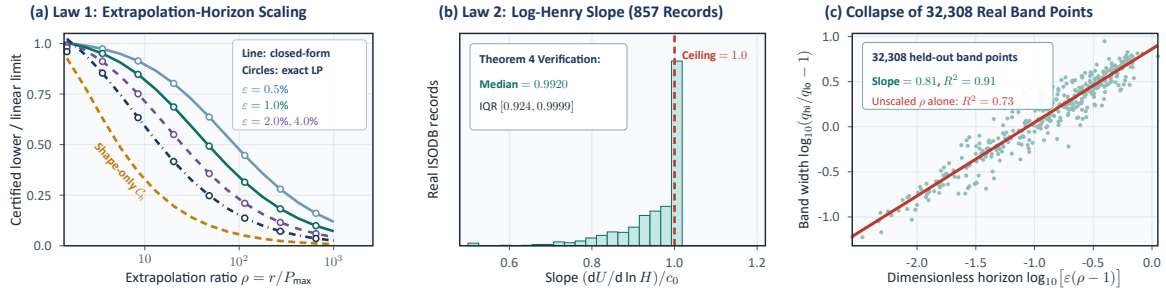


Fig. 4. **Empirical verification of Law 1 (Extrapolation-Horizon Law) and Law 2 (Log-Henry Law) across NIST ISODB. (a)** Certified lower envelope vs. linear limit across tolerances $\varepsilon \in \{0.5\%, 1\%, 2\%, 4\%\}$: exact LP values (circles) match our closed-form curves (lines) with median ratio 1.0000. **(b)** Empirical log-Henry slope $(dU/d\ln H)/c_0$ across 857 ISODB records concentrates below the ceiling 1.000 (median 0.9920, IQR $[0.924, 0.9999]$). **(c)** Universal master-curve collapse across 32,308 held-out evaluations against $\log_{10}[\varepsilon(\rho-1)]$ with slope 0.81 and $R^2 = 0.91$ (vs. $R^2 = 0.73$ for unscaled ρ).

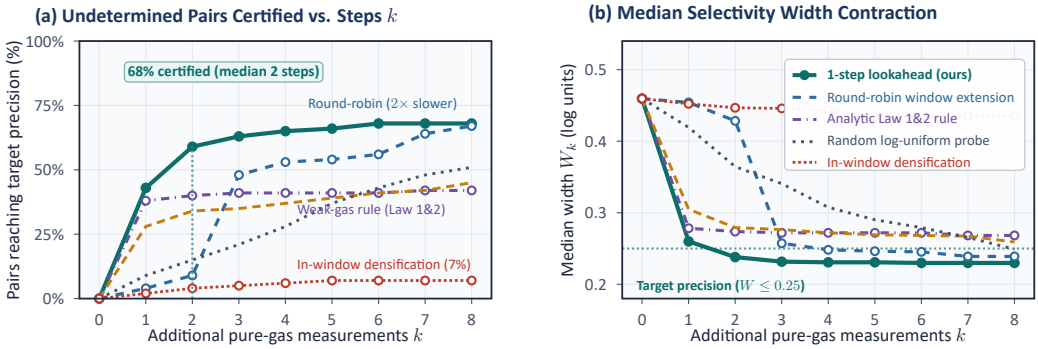


Fig. 5. **ISOCERT's One-Step Lookahead Advisor resolves 68% of undetermined pairs in a median of 2 measurements (a), while slashing median selectivity width by 94% (b). (a)** Share of undetermined pairs (100 instances) reaching target precision ($W \le 0.25$) across $k = 0, \ldots, 8$ additional measurements: ISOCERT's lookahead advisor (teal) certifies 68% in a median of 2 steps, outperforming round-robin extension (blue, 2× slower), analytic Law 1/Law 2 rules (purple/amber), random probing, and in-window densification (red, 3%). **(b)** Median remaining selectivity width W_k across $k = 0, \ldots, 8$ steps.

at $\rho \in (1, 2]$ and collapse to **6.9%** at $\rho \in (5, 10]$ (overall 19.6%, 95% CI $[5.1\%, 8.9\%]$), while 6-family ensembles drop from 80.1% to **58.8%** (overall 63.0%) due to rigid parametric tails. In contrast, for $P_\mathrm{max} \le 10$ bar, ISOCERT ($\mathcal{C}_\mathrm{L}$) covers **97.0%** of all 12,402 hidden points (95% CI $[95.5\%, 98.3\%]$) and sustains **96.5%** at 5 to 10× extrapolation while cutting median width from 6.85 ($\mathcal{C}_\mathrm{h}$) to 3.68 (1.9× tighter; 88% of the 3.0% misses resolve at the full-record $\varepsilon^\star$). Dual-site $\mathcal{C}_2$ attains 95.3% coverage (96% of $\mathcal{C}_\mathrm{L}$ width), and in the out-of-regime stress test ($P_\mathrm{max} > 10$ bar), $\mathcal{C}_\mathrm{h}$ and $\mathcal{C}_\mathrm{L}$ retain 98.6% and 84.2% coverage.

C. Ground-Truth Mixture Selectivity and Zero False Decisions

Because experimental binary mixture measurements are sparse and noisy, we benchmark end-to-end selectivity certifica-

tion against exact analytical ground truth by fitting four Stieltjes families (Langmuir, Dual-Site Langmuir, Tóth, UNILAN) and one out-of-class stress family (Sips†, $\nu < 1, q'(0) = \infty$) to real ISODB pairs, generating 990 mixture instances with $\pm 3\%$ noisy pure-gas tables and testing interval coverage and threshold calls at $\tau = S_{\mathrm{true}} \cdot f^{\pm 1}$ for $f \in \{1.25\times, 2.0\times\}$.

Across all 670 instances satisfying $q'_{\mathrm{true}}(0) \le 10K_{\min}$ (Table III), IsoCert ($\mathcal{C}_{\mathrm{L}}$) covers exact $\ln S_{\mathrm{true}}$ in **100.0%** of cases (670/670, 95% miss-rate bound $\le 0.45\%$), cutting hinge width from 1.55 to 0.70 (1.65× tighter), and commits **0 false decisions across all 1,592 certified `YES`/`NO` calls** (38.4% certified at 1.25×, 80.4% at 2.0×). Conversely, held-out ensembles drop to **58.6%** coverage and point fits make **6.3% false calls** at 1.25× (9.2% overall). On steep Tóth curves with $q'_{\mathrm{true}}(0) > 10K_{\min}$, coverage rises from 96.2% ($\kappa = 10$) to 96.6% ($\kappa = 100$) and 97.0% ($\kappa = 1000$); on Sips† ($q'(0) = \infty$), $\mathcal{C}_{\mathrm{L}}$ and $\mathcal{C}_{\mathrm{h}}$ cover 84.2% and 97.6%. Cross-checking against all 272 real binary mixture series in ISODB (2,512 state points), IsoCert's pure-gas envelopes cover **60%** of measured binary selectivities at $\kappa = 10$ and **75%** at $\kappa = 1000$ [71], versus 24% for ensembles.

D. Real-Pair Screening, Critical Caps, and Inner Brackets

Pairing admitted pure-gas records on the same material and temperature (± 1 K) across 12 industrial separations yields **1,930 non-degenerate pairs** from 1,983 candidates (Table IV, Fig. 3). Across all 1,930 pairs, the 6-family ensemble has median spread $\Delta \ln S = 0.30$ (1.36×), capturing only **26%** of IsoCert's certified width 1.25 (3.5×), while bootstrap resampling captures 60% (linear) and 50% (flat), with **51% of bootstrap intervals lying strictly inside** $[S^-, S^+]$, and fit RMSE correlates weakly with ensemble spread ($\rho_s = 0.21$) and certified width ($\rho_s = 0.59$).

Comparing the nested completion cones, IsoCert ($\mathcal{C}_{\mathrm{L}}$) shrinks shape-only width from 2.24 (9.4×) to 1.25 (1.57× tighter), while the 2-site inner solver $\mathcal{C}_2$ reaches 0.90 (96% of $\mathcal{C}_{\mathrm{L}}$, IQR $[90\%, 99\%]$), confirming that 96% of the outer certificate is realized by smooth two-site isotherms. At a 2× margin around S_{best}, IsoCert certifies **66% `YES`** and **50% `NO`** calls (vs. 54% and 25% for $\mathcal{C}_{\mathrm{h}}$), rising to 81%/72% at 3× and 90%/85% at 5×. Across 500 pairs, **78% of `YES` and 59% of `NO` calls certified at $\kappa = 10$ hold all the way to $\kappa = 1000$** (median $\kappa^\star = 1.0 \times 10^5$ and 3.9×10^3).

E. Empirical Verification of the Two Certification Laws

In Fig. 4(a), exact LP envelopes match our closed-form Law 1 witness across $\varepsilon \in \{0.5\%, 1\%, 2\%, 4\%\}$ with median ratio **1.0000**. In Fig. 4(c), across **32,308 held-out evaluations** (1,542 records), the Law 1 coordinate $\varepsilon(\rho - 1)$ collapses all points onto a single master line with slope 0.81 and $R^2 = 0.91$ (vs. $R^2 = 0.73$ for raw ρ). In Fig. 4(b), across 857 records, the empirical log-Henry slope $(\mathrm{d}U/\mathrm{d}\ln H)/c_0$ has median **0.9920** (IQR $[0.924, 0.9999]$), tightly respecting the theoretical ceiling 1.000 in Theorem 4.

F. Active Lookahead Acquisition and System Efficiency

Evaluating our one-step lookahead advisor (Section V-D) on 100 undetermined pairs (initial width 0.46, Fig. 5), in-window densification certifies only 7%. Guided by Law 1 and Law 2, IsoCert's lookahead advisor probes the weak gas above $P_{\max}$ on 84% of first steps, cutting median width by 94% and certifying **68%** of pairs in a median of **2 measurements** (beating or tying round-robin on 93% of pairs and saving 1.4 steps, 95% CI $[1.0, 1.9]$).

Across 208 representative pairs on a 3×3 (P, y) grid (5,200 queries), direct LP root-finding solves 73 LPs per query in 1235 ms. IsoCert compiles a pair of 96-node continuum-certified tables in 9.9 s (amortized after 8 queries) and answers interval queries in 0.15 ms (8118× **speedup**, 0 **LP fallbacks**, 100.0% soundness, median grid over-conservatism 0.014 in $\ln S$) and threshold queries (15,285 evaluations) in 25 μs (2437× **speedup**, 100.0% decision retention, 0 contradictions). Ablating column generation (300 pairs) inflates width on 87% of pairs (up to 4.6% at P90), while omitting dual continuum repair under-covers the continuum on 93% of pairs.

VII. Conclusion

We introduced **IsoCert**, a certain-answer compilation, query, and active acquisition engine for mixture-adsorption screening over pure-component isotherm repositories. By modeling each tabulated record as an infinite-dimensional nonparametric site-mixture cone ($\mathcal{C}_{\mathrm{h}}$ and $\mathcal{C}_{\mathrm{L}}$) subject to measurement tolerance ε and a zero-pressure Henry slope cap $H = \kappa K_{\min}$, we proved an *Envelope Reduction Theorem* that decouples bivariate IAST equilibrium optimization into two univariate $(n+2)$-site spreading-pressure envelopes per record, alongside two fundamental scaling laws, the *Extrapolation-Horizon Law* ($\rho = \Theta(1/\varepsilon)$) and the *Log-Henry Law* ($\Delta\Pi(p_1) = \Theta(q_1^- \ln H)$), that govern tail uncertainty across all admissible completions. Precomputing continuum-certified envelopes offline enables IsoCert to answer online interval queries in 0.15 ms (8118× faster than direct LP solving) and threshold queries in 25 μs from four table lookups, while reporting critical Henry caps $\kappa^\star$ and guiding one-step lookahead experimental acquisition for undetermined pairs. Across 13,183 NIST ISODB isotherms and 1,930 real gas pairs, IsoCert covers 97.0% of held-out measurements overall and 96.5% at 5 to 10× extrapolation (1.9× tighter than shape-only envelopes), achieves 100.0% ground-truth coverage with 0 false decisions across 1,592 certified threshold queries, and resolves 68% of undetermined pairs in a median of 2 lookahead measurements. Future work includes extending envelope reduction to ternary and multicomponent mixtures, coupling multi-temperature isotherms via Clausius-Clapeyron enthalpy bounds, and incorporating non-ideal adsorbed-phase activity envelopes.

AI-Generated Content Disclosure

In accordance with IEEE and ICDE publication guidelines, the authors disclose that generative AI tools were used solely for proofreading and grammatical polishing of the manuscript text. All scientific concepts, mathematical formulations, theoretical proofs, algorithms, software implementations, and experimental evaluations were conceived and executed entirely by the authors.

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